Beam Structured Signal Detector for HF Skywave Massive MIMO-OFDM Communications

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***Abstract*— In this paper, we investigate signal detection for HF skywave massive multiple-input multiple-output (MIMO) communications with orthogonal frequency division multiplexing (OFDM) modulation. We first introduce beam based channel models (BBCM) in the space domain at each subcarrier and in the space-frequency domain for all subcarriers. Based on the BBCM in the space domain, we propose a beam structured detector (BSD) for each subcarrier. Specifically, we prove that the space domain detector design can be transformed into that of a beam domain detector without sacrificing optimality, and the asymptotically optimal space domain detector is beam structured with a low-dimensional beam domain detector, thus significantly reducing the design and implementation complexities. Furthermore, we extend the BSD to the space-frequency domain based on the BBCM jointly for all subcarriers. The design of space-frequency domain detector is also converted to that of a low-dimensional beam domain detector, which enables a very efficient design and implementation of BSD. Simulation results demonstrate the low complexity and satisfactory performance of the proposed detectors.**

***Index Terms*— Massive MIMO-OFDM, HF skywave communications, signal detection, beam structured detector.**

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I. Introduction

FUTURE wireless network is demanded for seamless global service coverage including remote and rural areas [2], [3]. High frequency (HF) communications, typically in the 3 to 30 MHz frequency band, can achieve over-the-horizon transmission by ionospheric reflection and provide worldwide coverage with relatively low deployment cost [4], [5]. However, conventional point-to-point HF communications suffer from low data rates owing to the limited spectrum resource and complicated ionospheric conditions, which limits its application.

As a key technology for the fifth-generation cellular wireless communications and its future evolution, massive multiple-input multiple-output (MIMO) can significantly enhance the spectrum and energy efficiencies by deploying a large number of antennas at the base station (BS) and serving simultaneously a number of users in the same time-frequency resource. Recently, massive MIMO in combination with orthogonal frequency division multiplexing (OFDM) has been introduced into HF communications in [6], [7], and [8]. In [6], the asymptotically achievable sum-rates of minimum mean-squared error (MMSE) based uplink (UL) receiver and downlink (DL) precoder are derived by assuming accurate channel state information (CSI) at the BS. In [7], a triple-beam based channel model is established from the physical characteristic of HF skywave channels. Based on the channel model, a low-overhead pilot and a low-complexity channel estimators are designed. In [8], a low-complexity beam domain robust precoder with imperfect CSI is proposed for high Doppler scenarios in HF massive MIMO communications. It is demonstrated in [6], [7], and [8] that massive MIMO can provide significant performance gain for HF communications.

Besides channel estimation and precoding, the signal detection also plays an important role in massive MIMO communications and has been investigated in terrestrial cellular massive MIMO systems. Due to the increasing number of BS antennas and users, a maximum likelihood (ML) detector [9], even if can achieve the optimal performance, suffers from the unaffordable computational complexity [10], similarly to several other nonlinear detectors including local search based detectors [11], [12], [13] and sphere decoding [14], [15], [16]. To reduce the complexity, graph-based message passing detectors are proposed, such as belief propagation (BP) [17], approximate message passing (AMP) [18] and

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expectation propagation (EP) [19]. On the other hand, linear detectors are proposed, such as zero-forcing (ZF) detector and minimum mean-squared error (MMSE) detector [20], [21]. Nevertheless, they involve large-dimensional matrix inversion and multiplication in massive MIMO systems, which leads to the prohibitive computational complexity as the dimension gets large. Therefore, several approximate matrix inversion based methods, such as Neumann Series [22], Newton iteration [23], and polynomial expansion [24], can reduce the complexity. Moreover, many iterative techniques, such as the Gauss-Seidel [25], successive over-relaxation [26], conjugate-gradient [27] and recursive conjugate-gradient [28], can avoid a direct matrix inversion.

Note that the massive MIMO channel typically exhibits sparsity in the beam domain [29], which has been used to design low-complexity detectors recently [30], [31], [32], [33]. In these works, a discret Fourier transform (DFT) matrix is typically used to transform the channel from the space domain to the beam domain. In [30], the channel matrix is compressed by beam selection then a linear ZF detector with non-linear post-processing is developed for signal detection directly based on the compressed channel. The two-layer BP detector in [31] combines beam-domain and antenna-domain processing, where the beam-domain BP in the first layer is performed based on the compressed beam-domain channel and the antenna-domain BP in the second layer compensates the information loss due to the channel compression. The beam domain local MMSE detector in [32] uses a window to select a contiguous block of beams to construct the detector for each user. Moreover, in [33], the outputs in the log-likelihood ratio (LLR) domain of the overlapping local MMSE filters are selectively combined to further enhance the performance.

In this paper, we investigate signal detection for HF skywave massive MIMO-OFDM communications. We exploit the channel sparsity in the beam domain to design low-complexity beam structured detectors (BSDs). The main contributions are summarized as follows.

- We present beam based channel models (BBCMs) in the space domain at each subcarrier and in the space-frequency domain for all subcarriers, respectively, and reveal the sparsity of the space-beam domain channel and the space-frequency-beam domain channel for HF skywave massive MIMO-OFDM communications.
- Based on the space BBCM, we propose a BSD for each subcarrier by utilizing the channel sparsity in the space-beam domain. Specifically, we prove that the space domain detector can be designed by optimizing the beam domain detector without sacrificing optimality. Furthermore, the BSD with a low-dimensional beam domain detector is asymptotically optimal for a sufficiently large number of BS antennas, where the used beams are related to those that each user terminal (UT) occupies, thus reducing the design and implementation complexities. Moreover, a beam selection criterion for BSD in the space domain is also provided. The low-complexity design and implementation are presented by exploiting the specific structure of the space-beam matrix.
- Based on the space-frequency BBCM, we extend the BSD to the space-frequency domain for all subcarriers jointly by exploiting the space-frequency-beam domain channel sparsity. Specifically, we prove that the space-frequency domain detector design can be converted into the design of the beam domain detector while ensuring optimality. The BSD with a low-dimensional beam domain detector in the space-frequency domain is also asymptotically optimal when the number of BS antennas is sufficiently large. Moreover, we show that when certain conditions are satisfied, the used beams in the space-frequency domain only correspond to those that each UT occupies and the optimal beam domain detector is the scaled space-frequency-beam domain channel matrix, which leads to very low complexity.

The rest of the paper is organized as follows. In Section II, we present the channel models for HF skywave massive MIMO-OFDM communications. In Section III, the BSD in the space domain is proposed based on the space BBCM and the complexity is discussed. In Section IV, the BSD is extended to the space-frequency domain based on the space-frequency BBCM and the complexity is analyzed. Simulation results are provided in Section V and the paper is concluded in Section VI.

Notations: The uppercase (lowercase) boldface letters denote matrices (vectors). The superscripts $(\cdot)^*$, $(\cdot)^{\mathrm{T}}$, $(\cdot)^{\mathrm{H}}$, $(\cdot)^{\dagger}$ and $\mathrm{tr}\{\cdot\}$ denote the conjugate, transpose, conjugated-transpose, Moore-Penrose inverse and matrix trace operations, respectively. $\bar{\jmath} = \sqrt{-1}$ denotes the imaginary unit. $|\Psi|$ denotes the cardinality of set Ψ. $\lceil x \rceil$ denotes the smallest integer that is not less than x. $\circ$ and $\otimes$ denote the Hadamard and the Kronecker product operators, respectively. $[\mathbf{x}]_a$ and $[\mathbf{x}]_{a:b}$ denote the a-th element and elements a to b of vector $\mathbf{x}$, respectively. $[\mathbf{X}]_{a,b}$, $[\mathbf{X}]_{a,:}$, and $[\mathbf{X}]_{:,b}$ denote the (a,b)-th element, the a-th row, and the b-th colomn of $\mathbf{X}$, respectively. $\mathrm{vec}(\mathbf{X})$ denotes the vectorizing of a matrix $\mathbf{X}$. $\mathbb{E}\{\cdot\}$ denotes the statistical expectation. $\mathbf{1}_N$ denotes the all-one vector of length N. $\mathbf{I}_N$ denotes the identity matrix of dimension N. $\mathrm{diag}\{\mathbf{x}\}$ denotes the diagonal matrix with $\mathbf{x}$ along its main diagonal, and $\mathrm{diag}\{\mathbf{X}_1, \cdots, \mathbf{X}_N\}$ forms the block-diagonal matrix. For a vector $\mathbf{x} \in \mathbb{C}^{MN\times 1}$, $\mathrm{vdiag}\{\mathbf{x}\} = \mathrm{diag}\{[\mathbf{x}]_{0:M-1}, [\mathbf{x}]_{M:2M-1}, \cdots, [\mathbf{x}]_{(N-1)M:NM-1}\} \in \mathbb{C}^{MN\times N}$ denotes a vector-valued block-diagonal matrix, and $\mathrm{vcirc}\{\mathbf{x}\} = [\mathbf{x}, [[\mathbf{x}]^{\mathrm{T}}_{(N-1)M:NM-1}, [\mathbf{x}]^{\mathrm{T}}_{0:(N-1)M-1}]^{\mathrm{T}}, \cdots, [[\mathbf{x}]^{\mathrm{T}}_{M:NM-1}, [\mathbf{x}]^{\mathrm{T}}_{0:M-1}]^{\mathrm{T}}] \in \mathbb{C}^{MN\times N}$ denotes a vector-valued block-circulant matrix.

II. System Model

In this section, we present the system configuration of HF skywave massive MIMO-OFDM communications and introduce the BBCM in the space domain and in the space-frequency domain, respectively.

A. System Configuration

We consider a long-range over-the-horizon HF skywave massive MIMO-OFDM system as illustrated in Fig. 1, where

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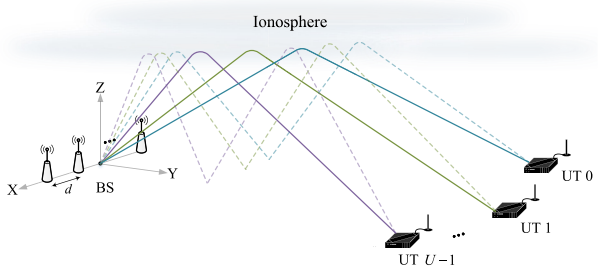


Fig. 1. Illustration of an HF skywave massive MIMO-OFDM system.

the BS deploys a uniform linear array (ULA) with M antennas and serves U single-antenna UTs.[1] Note that ULA configurations have been used for the HF skywave over-the-horizon MIMO radar, where arrays deploy hundreds of antennas, resulting in typically large antenna array aperture [35], [36], [37], which shows the feasibility also for the HF skywave massive MIMO communications.

Let N_{c} denote the number of subcarriers, where N_{v} subcarriers are used to transmit data with the index set $\mathcal{K} = \{k_0, k_1, \cdots, k_{N_{\mathrm{v}}-1}\}$. Moreover, we denote N_{g}, Δ_f, $T_{\mathrm{s}} = \frac{1}{N_{\mathrm{c}}\Delta_f}$, and $T_{\mathrm{sym}} = (N_{\mathrm{c}} + N_{\mathrm{g}})\, T_{\mathrm{s}}$ as the length of the cyclic prefix (CP), the subcarrier spacing, and the sampling interval, and the duration of an OFDM symbol with CP, respectively.

In HF skywave communications, the carrier frequency, f_{c}, needs to be adjusted according to different ionospheric conditions. Hence, the inter-antenna spacing, d, is set according to the highest used carrier frequency f_{o} instead of f_{c}, that is, $d = \lambda_{\mathrm{o}}/2$, where $\lambda_{\mathrm{o}} = c/f_{\mathrm{o}}$ is the wavelength, and c is the speed of light. Note that this differs from the conventional massive MIMO system, where the inter-antenna spacing d is typically set according to f_{c}.

B. Channel Model

During the UL transmission, the received analog baseband signal on the n-th OFDM symbol at the m-th antenna can be expressed as

$$\bar{y}_{m,n}(t) = \sum_{u=0}^{U-1} \int_{-\infty}^{\infty} \bar{h}_{u,m}(t,\tau)\, \bar{x}_{u,n}(t-\tau)\, d\tau + \bar{z}_{m,n}(t), \tag{1}$$

where $\bar{x}_{u,n}(t)$ is transmitted analog baseband signal on the n-th OFDM symbol of UT u, $m \in \{0, 1, \cdots, M-1\}$, $\bar{z}_{m,n}(t)$ is the additive white Gaussian noise and $\bar{h}_{u,m}(t,\tau)$ is the time-varying channel impulse response between UT u and the m-th antenna of the BS, which is

$$\bar{h}_{u,m}(t,\tau) = \sum_{p=0}^{P_u-1} \gamma_{u,p} e^{\bar{\jmath}2\pi\nu_{u,p}t} e^{-\bar{\jmath}2\pi f_{\mathrm{c}} m\Delta_\tau \Omega_{u,p}} \times \delta(\tau - \tau_{u,p} - m\Delta_\tau\Omega_{u,p}), \tag{2}$$

where P_u is the number of paths between the BS and UT u. In (2), $\gamma_{u,p}$ is the complex-valued gain of the p-th path of UT u and given by $\gamma_{u,p} = \beta_{u,p} e^{\bar{\jmath}\varphi_{u,p}}$, where $\beta_{u,p}$ and $\varphi_{u,p}$ are the gain and initial phase, respectively, and $\varphi_{u,p}$ is uniformly distributed over $[0, 2\pi)$. $\nu_{u,p}$ is the Doppler frequency, which is mainly caused by the ionosphere and the UT mobility. $\Omega_{u,p} \triangleq \sin\theta_{u,p}^{\mathrm{az}} \cos\theta_{u,p}^{\mathrm{el}}$ is the directional cosine, where $\theta_{u,p}^{\mathrm{az}}$ and $\theta_{u,p}^{\mathrm{el}}$ are the azimuth angle of arrival (AoA) and elevation AoA, respectively. $\tau_{u,p}$ is the p-th propagation path delay between UT u and the first antenna of the BS, and $\Delta_\tau = d/c$. Moreover, the propagation delay across the antenna array $m\Delta_\tau\Omega_{u,p}$ is also taken into account, which is known as the spatial-wideband effect and brought on by the increased bandwidth and equipped large-scale arrays [6], [7], [38].

We assume that the channel state keeps constant within an OFDM symbol and varies symbol by symbol due to the Doppler effect. The received data after OFDM demodulation at the k-th subcarrier on the n-th OFDM symbol at the m-th antenna is given by

$$y_{m,n,k} = \sum_{u=0}^{U-1} h_{u,m,n,k}^{\mathrm{SF}} x_{u,n,k} + z_{m,n,k}, \tag{3}$$

where $k \in \mathcal{K}$, $x_{u,n,k}$ denotes the transmitted data of UT u on the n-th OFDM symbol at the k-th subcarrier, $z_{m,n,k}$ is the additive white Gaussian noise with distribution $\mathcal{CN}(0, \sigma_{\mathrm{z}}^2)$, and $h_{u,m,n,k}^{\mathrm{SF}}$ is channel frequency response at the k-th subcarrier on the n-th OFDM symbol between UT u and the m-th antenna at the BS, which is given by

$$\begin{aligned} h_{u,m,n,k}^{\mathrm{SF}} &= \int \bar{h}_{u,m}(nT_{\mathrm{sym}}, \tau) e^{-\bar{\jmath}2\pi k\Delta_f \tau} d\tau \\ &= \sum_{p=0}^{P_u-1} \gamma_{u,p} e^{\bar{\jmath}2\pi\nu_{u,p} nT_{\mathrm{sym}}} e^{-\bar{\jmath}2\pi f_{\mathrm{c}} m\Delta_\tau\Omega_{u,p}} \\ &\quad \times e^{-\bar{\jmath}2\pi k\Delta_f \tau_{u,p}} e^{-\bar{\jmath}2\pi k\Delta_f m\Delta_\tau\Omega_{u,p}}. \end{aligned} \tag{4}$$

Let $\mathbf{h}_{u,n,k}^{\mathrm{S}} \in \mathbb{C}^{M\times 1}$ denote the space domain channel vector at the k-th subcarrier on the n-th OFDM symbol of UT u, with the m-th element, $h_{u,m,n,k}^{\mathrm{SF}}$, and define the space-frequency domain channel vector as $\mathbf{h}_{u,n}^{\mathrm{SF}} = \left[\left(\mathbf{h}_{u,n,k_0}^{\mathrm{S}}\right)^{\mathrm{T}}, \left(\mathbf{h}_{u,n,k_1}^{\mathrm{S}}\right)^{\mathrm{T}}, \cdots, \left(\mathbf{h}_{u,n,k_{N_{\mathrm{v}}-1}}^{\mathrm{S}}\right)^{\mathrm{T}}\right]^{\mathrm{T}}$. Therefore, we have

$$\mathbf{h}_{u,n,k}^{\mathrm{S}} = \sum_{p=0}^{P_u-1} \gamma_{u,p} e^{\bar{\jmath}2\pi\nu_{u,p} nT_{\mathrm{sym}}} e^{-\bar{\jmath}2\pi k\Delta_f \tau_{u,p}} \mathbf{v}_k(\Omega_{u,p}), \tag{5}$$

$$\mathbf{h}_{u,n}^{\mathrm{SF}} = \sum_{p=0}^{P_u-1} \gamma_{u,p} e^{\bar{\jmath}2\pi\nu_{u,p} nT_{\mathrm{sym}}} \mathbf{v}(\Omega_{u,p}) \circ (\mathbf{u}(\tau_{u,p}) \otimes \mathbf{1}_M), \tag{6}$$

where

$$\mathbf{v}_k(\Omega) \triangleq \left[1, e^{-\bar{\jmath}2\pi(f_{\mathrm{c}}+k\Delta_f)\Delta_\tau\Omega}, \cdots, e^{-\bar{\jmath}2\pi(f_{\mathrm{c}}+k\Delta_f)(M-1)\Delta_\tau\Omega}\right]^{\mathrm{T}} \in \mathbb{C}^{M\times 1}, \tag{7}$$

$$\mathbf{u}(\tau) \triangleq \left[e^{-\bar{\jmath}2\pi k_0\Delta_f\tau}, e^{-\bar{\jmath}2\pi k_1\Delta_f\tau}, \cdots, e^{-\bar{\jmath}2\pi k_{N_{\mathrm{v}}-1}\Delta_f\tau}\right]^{\mathrm{T}} \in \mathbb{C}^{N_{\mathrm{v}}\times 1}, \tag{8}$$

[1] Besides the long-range skywave communications considered in this paper, skywave can also be used for the so-called near vertical incidence skywave (NVIS) communications [34].

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are steering vectors in the space domain and the frequency domain, respectively, pointing towards directional cosine Ω and delay τ. Note that due to the spatial-wideband effect, the steering vectors in the space domain differ for various subcarriers, and we collect these into

$$\mathbf{v}\left(\Omega\right) \triangleq \left[\mathbf{v}_{k_0}(\Omega)^{\mathrm{T}}, \mathbf{v}_{k_1}(\Omega)^{\mathrm{T}}, \cdots, \mathbf{v}_{k_{N_{\mathrm{v}}-1}}(\Omega)^{\mathrm{T}}\right]^{\mathrm{T}} \in \mathbb{C}^{MN_{\mathrm{v}}\times 1}. \tag{9}$$

Note that the parameters $(\Omega_{u,p}, \tau_{u,p})$ of each path for each UT are limited within sets $\mathcal{B}_{\mathrm{an}} = \{\Omega \,|\Omega \in [-1, 1)\}$ and $\mathcal{B}_{\mathrm{de}} = \{\tau \,|\tau \in [0, \tau_{\max})\}$, respectively, where $\tau_{\max} \triangleq \frac{N_{\mathrm{g}}}{N_{\mathrm{c}}\Delta_f}$ is the maximum delay spread. We uniformly divide these sets into multiple disjoint subsets as

$$\mathcal{B}_{\mathrm{an}} = \bigcup_{n_{\mathrm{an}}=0}^{N_{\mathrm{an}}-1} \Lambda_{n_{\mathrm{an}}}^{\mathrm{an}}, \ \Lambda_{n_{\mathrm{an}}}^{\mathrm{an}} = \left[\frac{2n_{\mathrm{an}}-N_{\mathrm{an}}}{N_{\mathrm{an}}}, \frac{2n_{\mathrm{an}}-N_{\mathrm{an}}+2}{N_{\mathrm{an}}}\right), \tag{10a}$$

$$\mathcal{B}_{\mathrm{de}} = \bigcup_{n_{\mathrm{de}}=0}^{N_{\mathrm{de}}-1} \Lambda_{n_{\mathrm{de}}}^{\mathrm{de}}, \ \Lambda_{n_{\mathrm{de}}}^{\mathrm{de}} = \left[\frac{N_\tau n_{\mathrm{de}}}{N_{\mathrm{de}}N_{\mathrm{v}}\Delta_f}, \frac{N_\tau\left(n_{\mathrm{de}}+1\right)}{N_{\mathrm{de}}N_{\mathrm{v}}\Delta_f}\right), \tag{10b}$$

where $N_\tau \triangleq \frac{N_{\mathrm{v}}N_{\mathrm{g}}}{N_{\mathrm{c}}}$. The numbers of divided subsets for the directional cosine and the delay are $N_{\mathrm{an}} = \left\lceil F_{\mathrm{an}} M \frac{f_{\mathrm{c}}}{f_{\mathrm{o}}} \right\rceil$ and $N_{\mathrm{de}} = F_{\mathrm{de}}N_\tau$, respectively, where F_{an} and F_{de} are referred to as fine factors. Note that when N_{an} and N_{de} are relatively large, we can approximate the space domain steering vector $\mathbf{v}_k\left(\Omega_{u,p}\right)$ for $\Omega_{u,p} \in \Lambda_{n_{\mathrm{an}}}^{\mathrm{an}}$ as the sampled space domain steering vector $\mathbf{v}_k\left(\Omega_{n_{\mathrm{an}}}\right)$ and the frequency domain steering vector $\mathbf{u}\left(\tau_{u,p}\right)$ for $\tau_{u,p} \in \Lambda_{n_{\mathrm{de}}}^{\mathrm{de}}$ as the sampled frequency domain steering vector $\mathbf{u}\left(\tau_{n_{\mathrm{de}}}\right)$, where $\Omega_{n_{\mathrm{an}}} = \frac{2n_{\mathrm{an}}-N_{\mathrm{an}}}{N_{\mathrm{an}}}$ and $\tau_{n_{\mathrm{de}}} = \frac{N_\tau n_{\mathrm{de}}}{N_{\mathrm{de}}N_{\mathrm{v}}\Delta_f}$. Therefore, denoting $\Gamma_u^{\mathrm{S}} \triangleq \{\Omega_{u,0}, \Omega_{u,1}, \cdots, \Omega_{u,P_u-1}\}$, the space domain channel vector, $\mathbf{h}_{u,n,k}^{\mathrm{S}}$, becomes

$$\mathbf{h}_{u,n,k}^{\mathrm{S}} = \sum_{n_{\mathrm{an}}=0}^{N_{\mathrm{an}}-1} \tilde{h}_{u,n,k,n_{\mathrm{an}}}^{\mathrm{S}} \mathbf{v}_k\left(\Omega_{n_{\mathrm{an}}}\right), \tag{11}$$

$$\tilde{h}_{u,n,k,n_{\mathrm{an}}}^{\mathrm{S}} = \sum_{\Omega_{n_{\mathrm{an}}} \in \Gamma_u^{\mathrm{S}} \cap \Lambda_{n_{\mathrm{an}}}^{\mathrm{an}}} \gamma_{u,p} e^{\bar{\jmath}2\pi\nu_{u,p}nT_{\mathrm{sym}}} e^{-\bar{\jmath}2\pi k\Delta_f\tau_{u,p}}. \tag{12}$$

Similarly, denote $\Gamma_u^{\mathrm{SF}} \triangleq \left\{\left(\Omega_{u,0}, \tau_{u,0}\right), \left(\Omega_{u,1}, \tau_{u,1}\right), \cdots, \left(\Omega_{u,P_u-1}, \tau_{u,P_u-1}\right)\right\}$ and $\Lambda_{n_{\mathrm{an}},n_{\mathrm{de}}} \triangleq \Lambda_{n_{\mathrm{an}}}^{\mathrm{an}} \times \Lambda_{n_{\mathrm{de}}}^{\mathrm{de}}$, the space-frequency domain channel vector, $\mathbf{h}_{u,n}^{\mathrm{SF}}$, can be expressed as

$$\mathbf{h}_{u,n}^{\mathrm{SF}} = \sum_{n_{\mathrm{an}}=0}^{N_{\mathrm{an}}-1} \sum_{n_{\mathrm{de}}=0}^{N_{\mathrm{de}}-1} \tilde{h}_{u,n,n_{\mathrm{an}},n_{\mathrm{de}}}^{\mathrm{SF}} \mathbf{v}\left(\Omega_{n_{\mathrm{an}}}\right) \circ \left(\mathbf{u}\left(\tau_{n_{\mathrm{de}}}\right) \otimes \mathbf{1}_M\right), \tag{13}$$

$$\tilde{h}_{u,n,n_{\mathrm{an}},n_{\mathrm{de}}}^{\mathrm{SF}} = \sum_{\left(\Omega_{n_{\mathrm{an}}},\tau_{n_{\mathrm{de}}}\right) \in \Gamma_u^{\mathrm{SF}} \cap \Lambda_{n_{\mathrm{an}},n_{\mathrm{de}}}} \gamma_{u,p} e^{\bar{\jmath}2\pi\nu_{u,p}nT_{\mathrm{sym}}}. \tag{14}$$

Note that in (11) and (13), all the UTs share the same set of sampled space domain and frequency domain steering vectors. Each sampled space domain steering vector and each sampled frequency domain steering vector correspond to a physical beam in the space domain and in the frequency domain, respectively. Moreover, we denote the vector $\mathbf{v}\left(\Omega_{n_{\mathrm{an}}}\right) \circ \left(\mathbf{u}\left(\tau_{n_{\mathrm{de}}}\right) \otimes \mathbf{1}_M\right)$ in (13) that consists of the sampled space domain and the sampled frequency domain steering vectors as the sampled space-frequency domain steering vector, which corresponds to a physical beam in the space-frequency domain. Therefore, we refer to (11) and (13) as the BBCM in the space domain and in the space-frequency domain, respectively, where $\tilde{h}_{u,n,k,n_{\mathrm{an}}}^{\mathrm{S}}$ and $\tilde{h}_{u,n,n_{\mathrm{an}},n_{\mathrm{de}}}^{\mathrm{SF}}$ are the beam domain channel coefficient in space and in space-frequency, respectively.

Denote $\tilde{\mathbf{h}}_{u,n,k}^{\mathrm{S}}$ as the scaled space-beam domain channel with the n_{an}-th element, $\sqrt{M}\tilde{h}_{u,n,k,n_{\mathrm{an}}}^{\mathrm{S}}$, and define a space-beam matrix $\mathbf{V}_k = \frac{1}{\sqrt{M}}[\mathbf{v}_k(\Omega_0), \mathbf{v}_k(\Omega_1), \cdots, \mathbf{v}_k(\Omega_{N_{\mathrm{an}}-1})] \in \mathbb{C}^{M\times N_{\mathrm{an}}}$. Then (11) can be

$$\mathbf{h}_{u,n,k}^{\mathrm{S}} = \mathbf{V}_k \tilde{\mathbf{h}}_{u,n,k}^{\mathrm{S}}, \tag{15}$$

where each column of $\mathbf{V}_k$ corresponds to a space-beam. Thus the space-beam domain channel coupling vector of UT u, i.e., space-beam domain statistical CSI, is given by

$$\boldsymbol{\omega}_u^{\mathrm{S}} = \mathbb{E}\left\{\tilde{\mathbf{h}}_{u,n,k}^{\mathrm{S}} \circ \left(\tilde{\mathbf{h}}_{u,n,k}^{\mathrm{S}}\right)^*\right\}. \tag{16}$$

Similarly, denote $\tilde{\mathbf{h}}_{u,n}^{\mathrm{SF}}$ as the scaled space-frequency-beam domain channel with $(n_{\mathrm{de}}N_{\mathrm{an}} + n_{\mathrm{an}})$-th element $\sqrt{MN_{\mathrm{v}}}\tilde{h}_{u,n,n_{\mathrm{an}},n_{\mathrm{de}}}^{\mathrm{SF}}$ and define $\mathbf{U} = \frac{1}{\sqrt{N_{\mathrm{v}}}}[\mathbf{u}(\tau_0), \mathbf{u}(\tau_1), \cdots, \mathbf{u}(\tau_{N_{\mathrm{de}}-1})] \in \mathbb{C}^{N_{\mathrm{v}}\times N_{\mathrm{de}}}$, $\mathbf{V} = \mathrm{diag}\left\{\mathbf{V}_{k_0}, \cdots, \mathbf{V}_{k_{N_{\mathrm{v}}-1}}\right\}$. Then (13) becomes

$$\mathbf{h}_{u,n}^{\mathrm{SF}} = \mathbf{V}\left(\mathbf{U} \otimes \mathbf{I}_{N_{\mathrm{an}}}\right) \tilde{\mathbf{h}}_{u,n}^{\mathrm{SF}}, \tag{17}$$

where each column of $\mathbf{V}\left(\mathbf{U} \otimes \mathbf{I}_{N_{\mathrm{an}}}\right)$ corresponds to a space-frequency-beam. Thus the space-frequency-beam domain channel coupling vector of UT u, i.e., space-frequency-beam domain statistical CSI, is given by

$$\boldsymbol{\omega}_u^{\mathrm{SF}} = \mathbb{E}\left\{\tilde{\mathbf{h}}_{u,n}^{\mathrm{SF}} \circ \left(\tilde{\mathbf{h}}_{u,n}^{\mathrm{SF}}\right)^*\right\}. \tag{18}$$

In HF skywave massive MIMO communications, due to the small angular spread and limited number of propagation paths, the beam domain channels typically exhibit sparsity [7], [34], [39], [40], which means that most elements of the beam domain channels in space and in space-frequency are approximately zero. The ordered index set of non-zero elements of the space-beam domain channel for UT u is defined as $\mathcal{N}_u^{\mathrm{an}} = \left\{n_{\mathrm{an}} \middle| \left[\tilde{\mathbf{h}}_{u,n,k}^{\mathrm{S}}\right]_{n_{\mathrm{an}}} \neq 0\right\}$ with $\bar{N}_{\mathrm{an},u} = |\mathcal{N}_u^{\mathrm{an}}|$, where the order is the index increasing order. Define ordered set $\mathcal{N}^{\mathrm{an}} = \bigcup_{u=0}^{U-1} \mathcal{N}_u^{\mathrm{an}}$ with $\bar{N}_{\mathrm{an}} = |\mathcal{N}^{\mathrm{an}}|$. Denote $\bar{\mathbf{V}}_k = [\mathbf{V}_k]_{:,\mathcal{N}^{\mathrm{an}}} \in \mathbb{C}^{M\times \bar{N}_{\mathrm{an},u}}$ and $\bar{\mathbf{V}} = \mathrm{diag}\left\{\bar{\mathbf{V}}_{k_0}, \cdots, \bar{\mathbf{V}}_{k_{N_{\mathrm{v}}-1}}\right\}$, (15) becomes

$$\mathbf{h}_{u,n,k}^{\mathrm{S}} = \bar{\mathbf{V}}_k \bar{\mathbf{h}}_{u,n,k}^{\mathrm{S}}, \tag{19}$$

where $\bar{\mathbf{h}}_{u,n,k}^{\mathrm{S}} = \left[\tilde{\mathbf{h}}_{u,n,k}^{\mathrm{S}}\right]_{\mathcal{N}^{\mathrm{an}}}$, which shows that $\bar{N}_{\mathrm{an}}$ space-beams are sufficient to represent the space domain channel. Moreover, we define the

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ordered sets $\mathcal{N}_{u,n_{\rm an}}^{\rm de} = \{n_{\rm de} | [\tilde{\mathbf{h}}_{u,n}^{\rm SF}]_{n_{\rm de}N_{\rm an}+n_{\rm an}} \neq 0\}$, $\mathcal{N}_u^{\rm SFB} = \{n_{\rm SFB} | [\tilde{\mathbf{h}}_{u,n}^{\rm SF}]_{n_{\rm SFB}} \neq 0\}$ with $\bar{N}_{{\rm SFB},u} = |\mathcal{N}_u^{\rm SFB}| = \sum_{n_{\rm an}\in\mathcal{N}_u^{\rm an}} |\mathcal{N}_{u,n_{\rm an}}^{\rm de}|$, and $\mathcal{N}^{\rm SFB} = \bigcup_{u=0}^{U-1} \mathcal{N}_u^{\rm SFB}$ with $\bar{N}_{\rm SFB} = |\mathcal{N}^{\rm SFB}|$. Then (17) becomes

$$\begin{aligned}\mathbf{h}_{u,n}^{\rm SF} &= \bar{\mathbf{V}} \left(\mathbf{U} \otimes \mathbf{I}_{\bar{N}_{\rm an}}\right) \bar{\mathbf{h}}_{u,n}^{\rm SF} \\ &= \left[\mathbf{V} \left(\mathbf{U} \otimes \mathbf{I}_{N_{\rm an}}\right)\right]_{:,\mathcal{N}^{\rm SFB}} \left[\tilde{\mathbf{h}}_{u,n}^{\rm SF}\right]_{\mathcal{N}^{\rm SFB}}, \end{aligned} \tag{20}$$

where $\bar{\mathbf{h}}_{u,n}^{\rm SF} = [[\tilde{\mathbf{h}}_{u,n}^{\rm SF}]_{\mathcal{N}^{\rm an}}^{\rm T}, [\tilde{\mathbf{h}}_{u,n}^{\rm SF}]_{N_{\rm an}+\mathcal{N}^{\rm an}}^{\rm T}, \cdots, [\tilde{\mathbf{h}}_{u,n}^{\rm SF}]_{(N_{\rm de}-1)N_{\rm an}+\mathcal{N}^{\rm an}}^{\rm T}]^{\rm T} \in \mathbb{C}^{\bar{N}_{\rm an}N_{\rm de}\times 1}$. Typically the number of non-zero space-frequency-beams $\bar{N}_{\rm SFB}$ is much less than $\bar{N}_{\rm an}N_{\rm v}$ and $\bar{N}_{\rm an}N_{\rm de}$, which means that the channel has enhanced sparsity in the space-frequency-beam domain and $\bar{N}_{\rm SFB}$ space-frequency-beams are sufficient to represent the space-frequency domain channel. Note that non-zero beam indexes can also be obtained according to the statistical CSI as $\mathcal{N}_u^{\rm an} = \{n_{\rm an} | [\boldsymbol{\omega}_u^{\rm S}]_{n_{\rm an}} \neq 0\}$ and $\mathcal{N}_u^{\rm SFB} = \{n_{\rm SFB} | [\boldsymbol{\omega}_u^{\rm SF}]_{n_{\rm SFB}} \neq 0\}$. In this paper, we assume that the CSI is avaliable at the BS.[2]

III. BSD in the Space Domain

In this section, we investigate the UL detector at each subcarrier with the established space BBCM. We prove that the space domain detector design can be transformed into that of a beam domain detector without losing optimality. Moreover, when the number of BS antennas is sufficiently large, an asymptotically optimal space domain detector is introduced and is shown to be beam structured with a low-dimensional beam domain detector and thus can be designed and implemented with low complexity. Moreover, the beam selection criterion for the BSD is provided and the low-complexity design and implementation are presented.

A. Problem Formulation

We consider the UL transmission at the k-th subcarrier on the n-th OFDM symbol and omit index n in the subscript hereafter for brevity. The received signal in (3) is rewritten as

$$\mathbf{y}_k = \mathbf{H}_k^{\rm S}\mathbf{x}_k + \mathbf{z}_k, \tag{21}$$

where $\mathbf{y}_k = [y_{0,k}, y_{1,k}, \cdots, y_{M-1,k}]^{\rm T}$, $\mathbf{H}_k^{\rm S} = \left[\mathbf{h}_{0,k}^{\rm S}, \mathbf{h}_{1,k}^{\rm S}, \cdots, \mathbf{h}_{U-1,k}^{\rm S}\right]$, $\mathbf{x}_k = [x_{0,k}, x_{1,k}, \cdots, x_{U-1,k}]^{\rm T}$, and $\mathbf{z}_k \in \mathbb{C}^{M\times 1}$ is the additive white Gaussian noise with distribution $\mathcal{CN}\left(\mathbf{0}, \sigma_{\rm z}^2\mathbf{I}_M\right)$. We assume that $\mathbb{E}\left\{\mathbf{x}_k\mathbf{x}_k^{\rm H}\right\} = \sigma_{\rm x}^2\mathbf{I}_U$, where $\sigma_{\rm x}^2$ is the average transmit power of each UT.

Consider a linear detector $\mathbf{w}_{u,k} \in \mathbb{C}^{M\times 1}$ for UT u and define $\mathbf{W}_k = [\mathbf{w}_{0,k}, \mathbf{w}_{1,k}, \cdots, \mathbf{w}_{U-1,k}] \in \mathbb{C}^{M\times U}$. Then the mean-squared error (MSE) of the detection result is given by

$$\mathrm{MSE}_k = \mathbb{E}\left\{\left\|\mathbf{x}_k - \mathbf{W}_k^{\rm H}\mathbf{y}_k\right\|^2\right\}. \tag{22}$$

[2]The CSI can be obtained by channel estimation combined with the channel prediction approach in [7].

Thus the conventional space domain MMSE detector is

$$\mathbf{W}_k^{\rm mmse} = \arg\min_{\mathbf{W}_k} \mathrm{MSE}_k = \mathbf{H}_k^{\rm S}\left(\left(\mathbf{H}_k^{\rm S}\right)^{\rm H}\mathbf{H}_k^{\rm S} + \frac{\sigma_{\rm z}^2}{\sigma_{\rm x}^2}\mathbf{I}_U\right)^{-1}, \tag{23}$$

and the detection result is

$$\hat{\mathbf{x}}_k = (\mathbf{W}_k^{\rm mmse})^{\rm H}\mathbf{y}_k. \tag{24}$$

Thus the design and implementation complexities of the MMSE detector are $\mathcal{O}\left(MU^2 + U^3\right)$ and $\mathcal{O}\left(MU\right)$ according to (23) and (24), respectively. In HF skywave massive MIMO-OFDM communications, with the increased number of BS antennas and UTs, the large M and U will make the complexity of the MMSE detector very high due to the large-dimensional matrix inversion and multiplication. However, the angular spread of the HF skywave channel is typically small [7], [34], [39], [40], which means that the channel has the significant sparsity in the beam domain and the channel energy of each UT is concentrated on a few non-zero beams. Therefore, this motivates us to utilize such a sparsity in the beam domain to design a structured detector, which allows much reduced design and implementation complexities.

B. BSD in the Space Domain

We consider the UL detector design for UT u, whose MSE for signal detection is

$$\begin{aligned}\mathrm{MSE}_{u,k} &= \mathbb{E}\left\{\left\|x_{u,k} - \mathbf{w}_{u,k}^{\rm H}\mathbf{y}_k\right\|^2\right\} \\ &= \sigma_{\rm x}^2 - \sigma_{\rm x}^2\mathbf{w}_{u,k}^{\rm H}\mathbf{h}_{u,k}^{\rm S} - \sigma_{\rm x}^2\left(\mathbf{h}_{u,k}^{\rm S}\right)^{\rm H}\mathbf{w}_{u,k} \\ &\quad + \mathbf{w}_{u,k}^{\rm H}\left(\sigma_{\rm x}^2\sum_{u'=0}^{U-1}\mathbf{h}_{u',k}^{\rm S}\left(\mathbf{h}_{u',k}^{\rm S}\right)^{\rm H} + \sigma_{\rm z}^2\mathbf{I}\right)\mathbf{w}_{u,k}. \end{aligned} \tag{25}$$

Then the design of detector $\mathbf{w}_{u,k}$ for UT u can be formulated as the following problem

$$\mathbf{w}_{u,k}^{\rm op} = \arg\min_{\mathbf{w}_{u,k}} \mathrm{MSE}_{u,k}. \tag{26}$$

Motivated by the sparse space BBCM (19), we rewrite $\mathrm{MSE}_{u,k}$ as

$$\begin{aligned}\mathrm{MSE}_{u,k} &= \sigma_{\rm x}^2 - \sigma_{\rm x}^2\mathbf{w}_{u,k}^{\rm H}\bar{\mathbf{V}}_k\bar{\mathbf{h}}_{u,k}^{\rm S} - \sigma_{\rm x}^2\left(\bar{\mathbf{V}}_k\bar{\mathbf{h}}_{u,k}^{\rm S}\right)^{\rm H}\mathbf{w}_{u,k} \\ &\quad + \mathbf{w}_{u,k}^{\rm H}\left(\sigma_{\rm x}^2\sum_{u'=0}^{U-1}\bar{\mathbf{V}}_k\bar{\mathbf{h}}_{u',k}^{\rm S}\left(\bar{\mathbf{V}}_k\bar{\mathbf{h}}_{u',k}^{\rm S}\right)^{\rm H} + \sigma_{\rm z}^2\mathbf{I}\right)\mathbf{w}_{u,k}. \end{aligned} \tag{27}$$

In (27), the space-beam domain channel $\bar{\mathbf{h}}_{u,k}^{\rm S}$ instead of the space domain channel $\mathbf{h}_{u,k}^{\rm S}$ is involved. Then we have the following theorem, proved in Appendix A.

Theorem 1: The optimal solution of problem in (26) can be obtained as

$$\mathbf{w}_{u,k}^{\rm op} = \bar{\mathbf{V}}_k\left(\bar{\mathbf{V}}_k^{\rm H}\bar{\mathbf{V}}_k\right)^{\dagger}\tilde{\mathbf{w}}_{u,k}^{\rm op}, \tag{28}$$

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where

$$\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}} = \underset{\tilde{\mathbf{w}}_{u,k}}{\operatorname{argmin}} \Big\{ \sigma_{\mathrm{x}}^2 - \sigma_{\mathrm{x}}^2 \tilde{\mathbf{w}}_{u,k}^{\mathrm{H}} \bar{\mathbf{h}}_{u,k}^{\mathrm{S}} - \sigma_{\mathrm{x}}^2 \big(\bar{\mathbf{h}}_{u,k}^{\mathrm{S}}\big)^{\mathrm{H}} \tilde{\mathbf{w}}_{u,k} + \tilde{\mathbf{w}}_{u,k}^{\mathrm{H}} \Big(\sigma_{\mathrm{x}}^2 \sum_{u'=0}^{U-1} \bar{\mathbf{h}}_{u',k}^{\mathrm{S}} \big(\bar{\mathbf{h}}_{u',k}^{\mathrm{S}}\big)^{\mathrm{H}} + \sigma_{\mathrm{z}}^2 \big(\bar{\mathbf{V}}_k^{\mathrm{H}} \bar{\mathbf{V}}_k\big)^{\dagger} \Big) \tilde{\mathbf{w}}_{u,k} \Big\}$$
$$\text{s.t.} \quad \Big(\bar{\mathbf{V}}_k^{\mathrm{H}} \big(\bar{\mathbf{V}}_k^{\mathrm{H}}\big)^{\dagger} - \mathbf{I} \Big) \tilde{\mathbf{w}}_{u,k} = \mathbf{0}. \tag{29}$$

Theorem 1 provides an optimal detector structure. In the optimization problem (29), we can turn to optimizing $\tilde{\mathbf{w}}_{u,k}$ instead of $\mathbf{w}_{u,k}$ to minimize the MSE. When we obtain $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}}$ by solving the optimization problem in (29), we can directly construct detector $\mathbf{w}_{u,k}^{\mathrm{op}}$ according to the relation (28). Moreover, we assume that the number of UTs is limited and the directional cosines of the paths of UTs are distinct [6]. Thus when $M \to \infty$, the columns of $\bar{\mathbf{V}}_k$ are asymptotically orthogonal and it can be checked that $\lim_{M\to\infty} \bar{\mathbf{V}}_k^H \big(\bar{\mathbf{V}}_k^H\big)^{\dagger} - \mathbf{I} = \mathbf{0}$ and $\lim_{M\to\infty} \big(\bar{\mathbf{V}}_k^{\mathrm{H}} \bar{\mathbf{V}}_k\big)^{\dagger} = \mathbf{I}$. In this case, the optimal solution in (28) becomes

$$\mathbf{w}_{u,k}^{\mathrm{op},\infty} = \bar{\mathbf{V}}_k \tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}, \tag{30}$$

where

$$\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty} = \underset{\tilde{\mathbf{w}}_{u,k}}{\operatorname{argmin}} \Big\{ \sigma_{\mathrm{x}}^2 - \sigma_{\mathrm{x}}^2 \tilde{\mathbf{w}}_{u,k}^{\mathrm{H}} \bar{\mathbf{h}}_{u,k}^{\mathrm{S}} - \sigma_{\mathrm{x}}^2 \big(\bar{\mathbf{h}}_{u,k}^{\mathrm{S}}\big)^{\mathrm{H}} \tilde{\mathbf{w}}_{u,k} + \tilde{\mathbf{w}}_{u,k}^{\mathrm{H}} \Big(\sigma_{\mathrm{x}}^2 \sum_{u'=0}^{U-1} \bar{\mathbf{h}}_{u',k}^{\mathrm{S}} \big(\bar{\mathbf{h}}_{u',k}^{\mathrm{S}}\big)^{\mathrm{H}} + \sigma_{\mathrm{z}}^2 \mathbf{I} \Big) \tilde{\mathbf{w}}_{u,k} \Big\}. \tag{31}$$

The relation in (30) provides an asymptotically optimal detector structure, which is much simpler than (28). Meanwhile, the optimization problem in (31) also has a simpler form. Since only the space-beam domain channels are involved in (31) and the structure in (30) is very similar to that of space BBCM (19), we refer to $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}$ as the beam domain detector. The resulting space domain detector, $\mathbf{w}_{u,k}^{\mathrm{op},\infty}$, is referred to as the BSD in the space domain.

In the optimization problem in (31), the dimension of optimization variable $\tilde{\mathbf{w}}_{u,k}$ is still large since it considers non-zero beams of all UTs and the search for $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}$ is still computationally expensive. Thus we have the following theorem, proved in Appendix B, which can significantly reduce the dimension of $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}$.

Theorem 2: When $\forall u, u', \mathcal{N}_u^{\mathrm{an}} \cap \mathcal{N}_{u'}^{\mathrm{an}} = \emptyset$, the optimal solution of (31) satisfies

$$\Big[\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}\Big]_{n_{\mathrm{an}}} = 0, \quad \text{for } [\mathcal{N}^{\mathrm{an}}]_{n_{\mathrm{an}}} \notin \mathcal{N}_u^{\mathrm{an}}. \tag{32}$$

From Theorem 2, when there are no overlapping beam domain channels for any two UTs, the elements of $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}$ are non-zero only in the positions corresponding to the non-zero beams that UT u occupies, for a sufficiently large M. Due to the sparsity of the space-beam domain channel for HF skywave massive MIMO communications, the number of non-zero beams of UT u, $\bar{N}_{\mathrm{an},u}$, is much smaller than the number of the total beams N_{an} and antennas M. Thus we focus on the non-zero elements of $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}$ and the dimensions of the optimization variable can be much smaller. Therefore, the optimal solution in (30) can be rewritten as

$$\mathbf{w}_{u,k}^{\mathrm{op},\infty} = \mathbf{V}_k \mathbf{S}_u \bar{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}, \tag{33}$$

where

$$\begin{aligned} \bar{\mathbf{w}}_{u,k}^{\mathrm{op},\infty} &= \underset{\bar{\mathbf{w}}_{u,k}}{\operatorname{argmin}} \Big\{ \sigma_{\mathrm{x}}^2 - \sigma_{\mathrm{x}}^2 \bar{\mathbf{w}}_{u,k}^{\mathrm{H}} \mathbf{S}_u^{\mathrm{H}} \tilde{\mathbf{h}}_{u,k}^{\mathrm{S}} - \sigma_{\mathrm{x}}^2 \Big(\tilde{\mathbf{h}}_{u,k}^{\mathrm{S}}\Big)^{\mathrm{H}} \mathbf{S}_u \bar{\mathbf{w}}_{u,k} \\ &\quad + \bar{\mathbf{w}}_{u,k}^{\mathrm{H}} \Big(\sigma_{\mathrm{x}}^2 \sum_{u'=0}^{U-1} \mathbf{S}_u^{\mathrm{H}} \tilde{\mathbf{h}}_{u',k}^{\mathrm{S}} \Big(\tilde{\mathbf{h}}_{u',k}^{\mathrm{S}}\Big)^{\mathrm{H}} \mathbf{S}_u + \sigma_{\mathrm{z}}^2 \mathbf{I} \Big) \bar{\mathbf{w}}_{u,k} \Big\} \\ &= \Big(\sum_{u'=0}^{U-1} \mathbf{S}_u^{\mathrm{H}} \tilde{\mathbf{h}}_{u',k}^{\mathrm{S}} \Big(\tilde{\mathbf{h}}_{u',k}^{\mathrm{S}}\Big)^{\mathrm{H}} \mathbf{S}_u + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2} \mathbf{I} \Big)^{-1} \mathbf{S}_u^{\mathrm{H}} \tilde{\mathbf{h}}_{u,k}^{\mathrm{S}}, \end{aligned} \tag{34}$$

and $\mathbf{S}_u \in \mathbb{R}^{N_{\mathrm{an}} \times \bar{N}_{\mathrm{an},u}}$ is the beam selection martrix with

$$[\mathbf{S}_u]_{i,j} = \begin{cases} 1, & i = [\mathcal{N}_u^{\mathrm{an}}]_j \\ 0, & \text{otherwise}, \end{cases} \tag{35}$$

and $\bar{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}$ is the low-dimensional beam domain detector corresponding to the non-zeros elements of $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}$. For brevity, we denote the BSD in the space domain in (33) with the beam index set $\mathcal{N}_u^{\mathrm{an}}$ as $\mathbf{w}_{u,k}^{\mathrm{op},\infty}(\mathcal{N}_u^{\mathrm{an}})$.

When the number of UTs becomes larger, the condition that the beam domain channels for any two UTs are non-overlapping, however, may not be well satisfied. In this case, more beams need to be considered to improve the detection performance. For this purpose, we have the following Theorem, proved in Appendix C, where $\mathrm{MSE}_{u,k}\big(\mathbf{w}_{u,k}^{\mathrm{op},\infty}(\mathcal{N}_{u,1}^{\mathrm{an}})\big)$ and $\mathrm{MSE}_{u,k}\big(\mathbf{w}_{u,k}^{\mathrm{op},\infty}(\mathcal{N}_{u,2}^{\mathrm{an}})\big)$ refer to the MSEs of detectors, $\mathbf{w}_{u,k}^{\mathrm{op},\infty}(\mathcal{N}_{u,1}^{\mathrm{an}})$ and $\mathbf{w}_{u,k}^{\mathrm{op},\infty}(\mathcal{N}_{u,2}^{\mathrm{an}})$, respectively.

Theorem 3: For a given beam index set $\mathcal{N}_{u,1}^{\mathrm{an}} \supseteq \mathcal{N}_u^{\mathrm{an}}$, define a new beam index set $\mathcal{N}_{u,2}^{\mathrm{an}} = \mathcal{N}_{u,1}^{\mathrm{an}} \cup \{\tilde{n}_{\mathrm{an}}\}$ where $\tilde{n}_{\mathrm{an}} \notin \mathcal{N}_{u,1}^{\mathrm{an}}$. When $M \to \infty$, we have

$$\mathrm{MSE}_{u,k}\big(\mathbf{w}_{u,k}^{\mathrm{op},\infty}(\mathcal{N}_{u,1}^{\mathrm{an}})\big) \geq \mathrm{MSE}_{u,k}\big(\mathbf{w}_{u,k}^{\mathrm{op},\infty}(\mathcal{N}_{u,2}^{\mathrm{an}})\big), \tag{36}$$

where equality holds provided that

$$\big[\tilde{\mathbf{H}}_k^{\mathrm{S}}\big]_{\mathcal{N}_{u,1}^{\mathrm{an}},:} \Big[\tilde{\mathbf{H}}_k^{\mathrm{S}}\Big]_{\tilde{n}_{\mathrm{an}},:}^{\mathrm{H}} = \mathbf{0}, \tag{37}$$

where $\tilde{\mathbf{H}}_k^{\mathrm{S}} = \big[\tilde{\mathbf{h}}_{0,k}^{\mathrm{S}}, \cdots, \tilde{\mathbf{h}}_{U-1,k}^{\mathrm{S}}\big]$.

From Theorem 3, when the beam domain channel of UT u overlaps with those of other UTs, it is not optimal to use only the beams occupied by UT u, in the BSD design, and more beams need to be used for the detector design. Moreover, Theorem 3 provides a criterion for the selection of the additional beams. The beams that satisfy the condition (37) should not be selected since it will not improve the performance of the detector, while the beams that satisfy the condition $\big[\tilde{\mathbf{H}}_k^{\mathrm{S}}\big]_{\mathcal{N}_{u,1}^{\mathrm{an}},:} \big[\tilde{\mathbf{H}}_k^{\mathrm{S}}\big]_{\tilde{n}_{\mathrm{an}},:}^{\mathrm{H}} \neq \mathbf{0}$ should be selected to improve the performance of the BSD. Here, we illustrate it in Fig 2 as an example. As shown in Fig 2, when the beam set of UT 2 overlaps with that of UTs 1 and 3, we can select another two beams in addition to the beams occupied by UT 2 as shown by the solid line box to improve the performance of the detector. Moreover, we can add another two beams as shown in dotted line box to further reduce the MSE and improve

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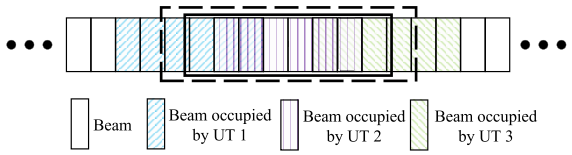


Fig. 2. Illustration of Theorem 3.

the performance. According to Theorem 3, if we want to achieve the optimal detector performance, all the beams of the UT whose beam index set overlaps with the already selected beam index set should be selected. However, as shown in the simulation section, satisfactory performance can be achieved by selecting only a few beams around the UT in addition to the beams it occupies.

Although the number of BS antennas in practical systems is finite, the above discussion suggests that the BSD still can be used to achieve near-optimal performance and low complexity when the number of BS antennas is large. Therefore, we constrain the space domain detector of UT u to have the structure $\mathbf{w}_{u,k} = \mathbf{V}_k \hat{\mathbf{S}}_u \hat{\mathbf{w}}_{u,k}$, where the definition of the beam selection matrix $\hat{\mathbf{S}}_u$ is similar to $\mathbf{S}_u$ except that the ordered selected beam index set, $\mathcal{N}_u^{\mathrm{an}}$, is replaced with $\hat{\mathcal{N}}_u^{\mathrm{an}} \supseteq \mathcal{N}_u^{\mathrm{an}}$ and $\left|\hat{\mathcal{N}}_u^{\mathrm{an}}\right| = \hat{N}_{\mathrm{an},u}$. Thus we turn to optimizing $\hat{\mathbf{w}}_{u,k}$, whose optimal solution in terms of MSE minimization is

$$\begin{aligned}\hat{\mathbf{w}}_{u,k}^{\mathrm{B}} &= \arg\min_{\hat{\mathbf{w}}_{u,k}} \mathbb{E}\left\{\left\|x_{u,k} - \hat{\mathbf{w}}_{u,k}^{\mathrm{H}} \hat{\mathbf{S}}_u^{\mathrm{H}} \mathbf{V}_k^{\mathrm{H}} \mathbf{y}_k\right\|^2\right\} \\ &= \left(\sum_{u'=0}^{U-1} \mathbf{P}_{u,u',k} \hat{\mathbf{h}}_{u',k}^{\mathrm{S}} \left(\hat{\mathbf{h}}_{u',k}^{\mathrm{S}}\right)^{\mathrm{H}} \mathbf{P}_{u',u,k} + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2} \mathbf{P}_{u,u,k}\right)^{-1} \\ &\quad \times \mathbf{P}_{u,u,k} \hat{\mathbf{h}}_{u,k}^{\mathrm{S}}, \end{aligned} \tag{38}$$

where $\mathbf{P}_{u,u',k} \triangleq \hat{\mathbf{S}}_u^{\mathrm{H}} \mathbf{V}_k^{\mathrm{H}} \mathbf{V}_k \hat{\mathbf{S}}_{u'} \in \mathbb{C}^{\hat{N}_{\mathrm{an},u} \times \hat{N}_{\mathrm{an},u'}}$ and $\hat{\mathbf{h}}_{u,k}^{\mathrm{S}} \in \mathbb{C}^{\hat{N}_{\mathrm{an},u} \times 1}$ is the low-dimensional space-beam domain channel corresponding to the beam index set $\hat{\mathcal{N}}_u^{\mathrm{an}}$. Then the space domain detector is

$$\mathbf{w}_{u,k}^{\mathrm{B}} = \mathbf{V}_k \hat{\mathbf{S}}_u \hat{\mathbf{w}}_{u,k}^{\mathrm{B}}, \tag{39}$$

and the detection result is

$$\hat{x}_u = \left(\mathbf{w}_{u,k}^{\mathrm{B}}\right)^{\mathrm{H}} \mathbf{y} = \left(\hat{\mathbf{w}}_{u,k}^{\mathrm{B}}\right)^{\mathrm{H}} \hat{\mathbf{S}}_u^{\mathrm{H}} \mathbf{V}_k^{\mathrm{H}} \mathbf{y}_k. \tag{40}$$

Note that the beam index set, $\hat{\mathcal{N}}_u^{\mathrm{an}}$, can be flexibly selected to trade off the performance and complexity. Moreover, when M tends to infinity and $\hat{\mathcal{N}}_u^{\mathrm{an}} = \mathcal{N}_u^{\mathrm{an}}$, (38) will become (34).

C. Complexity Analysis

We analyze the complexity of the proposed BSD in the space domain in this subsection. For simplicity, we define $\hat{N}_{\mathrm{an}}^{\mathrm{ave}} = \frac{1}{U} \sum_{u=0}^{U-1} \hat{N}_{\mathrm{an},u}$ and $\hat{N}_{\mathrm{an}}^{\max} = \max\left(\hat{N}_{\mathrm{an},u}\right)$. We first analyze the design complexity of the detector, which comes from (38). We can find that only low-dimensional space-beam domain channels $\hat{\mathbf{h}}_{0,k}^{\mathrm{S}}, \cdots, \hat{\mathbf{h}}_{U-1,k}^{\mathrm{S}}$ are involved in (38), and the dimension of matrix inversion is greatly reduced and depends on the number of selected beams. Moreover, the matrix $\mathbf{P}_{u,u',k}$ can be pre-calculated offline and used for a relatively long period since it is related to the non-zero beam indexes, which rely on the statistical CSI and vary slowly [7], [38], [41], [42]. Therefore, the design complexity of the proposed detector in terms of complex multiplication number is $\mathcal{O}\left(\left(\hat{N}_{\mathrm{an}}^{\max}\right)^2 U^2 + \left(\hat{N}_{\mathrm{an}}^{\max}\right)^3 U\right)$. Due to the significant sparsity in the beam domain of HF skywave massive MIMO-OFDM channels, $\hat{N}_{\mathrm{an}}^{\max}$ is small in general, which implies that the design complexity of the proposed detector is much less than the design complexity $\mathcal{O}\left(MU^2 + U^3\right)$ of the space domain MMSE detector in (23).

We then discuss the implementation complexity with the proposed detector, which comes from (40). The computation of (40) can be divided into two steps: $\tilde{\mathbf{y}}_k = \mathbf{V}_k^{\mathrm{H}} \mathbf{y}_k$ and $\hat{x}_u = \left(\hat{\mathbf{w}}_{u,k}^{\mathrm{B}}\right)^{\mathrm{H}} \hat{\mathbf{S}}_u^{\mathrm{H}} \tilde{\mathbf{y}}_k$. The first step can be regarded as transforming the space domain received signal $\mathbf{y}_k$ to the beam domain $\tilde{\mathbf{y}}_k$. It is not related to the specific UT and needs to be calculated only once for all UTs. The second step is to apply the designed low-dimensional beam domain detector $\hat{\mathbf{w}}_{u,k}^{\mathrm{B}}$ to the extracted beam domain received signal with the beam selection matrix $\hat{\mathbf{S}}_u$. Therefore, the complexity mainly comes from the first step, whose complexity is $\mathcal{O}\left(MN_{\mathrm{an}}\right)$ if we directly compute it. Such complexity is still high since the number of BS antennas is large. Note that $\mathbf{V}_k$ can be treated as a chirp z-transform (CZT) matrix and rewritten as [43]

$$\begin{aligned}\mathbf{V}_k = &\operatorname{diag}\left\{\boldsymbol{\xi}_{W_{\mathrm{S},k},N_{\mathrm{an}},0,M}\right\} \mathbf{F}_{N_{(\mathrm{S})} \times M}^{\mathrm{H}} \\ &\times \operatorname{diag}\left\{\tilde{\boldsymbol{\xi}}_{W_{\mathrm{S},k},M,N_{\mathrm{an}},N_{(\mathrm{S})}}\right\} \\ &\times \mathbf{F}_{N_{(\mathrm{S})} \times N_{\mathrm{an}}} \operatorname{diag}\left\{\boldsymbol{\xi}_{W_{\mathrm{S},k},0,0,N_{\mathrm{an}}}\right\}, \end{aligned} \tag{41}$$

where $\mathbf{F}_N$ denotes the N-dimensional unitary DFT matrix, $\mathbf{F}_{N \times G}$ denotes the matrix composed of the first G ($\leq N$) columns of $\mathbf{F}_N$, $N_{(\mathrm{S})}$ is an integer greater than or equal to $M + N_{\mathrm{an}} - 1$. $W_{\mathrm{S},k} \triangleq e^{-\bar{\jmath}\frac{4\pi\left(f_{\mathrm{c}} + k\Delta_f\right)\Delta_\tau}{N_{\mathrm{an}}}}$, $\boldsymbol{\xi}_{W,\alpha_1,\alpha_2,N} \in \mathbb{C}^{N \times 1}$ with $\left[\boldsymbol{\xi}_{W,\alpha_1,\alpha_2,N}\right]_i = W^{\left(i^2 - \alpha_1 i - \alpha_2\right)/2}$, and $\tilde{\boldsymbol{\xi}}_{W,N_1,N_2,N} \triangleq \sqrt{N}\mathbf{F}_N\left[\boldsymbol{\xi}_{W,0,0,N_1}^{\mathrm{H}}, \mathbf{0}, \boldsymbol{\xi}_{W,2(N_2-1),-(N_2-1)^2,N_2-1}^{\mathrm{H}}\right]^{\mathrm{T}} \in \mathbb{C}^{N \times 1}$. Thus the multiplication $\mathbf{V}_k^{\mathrm{H}} \mathbf{y}$ only consists of the product of diagonal matrices or DFT matrices and the vector. Moreover, the product of DFT matrices and the vector can be implemented efficiently by fast Fourier transform (FFT). In conclusion, the implementation complexity of the proposed detector in terms of complex multiplication number is $\mathcal{O}\left(N_{(\mathrm{S})} \log_2 N_{(\mathrm{S})} + \hat{N}_{\mathrm{an}}^{\mathrm{ave}} U\right)$ while the implementation complexity of the space domain MMSE detector is $\mathcal{O}\left(MU\right)$ according to (24).

IV. BSD in the Space-Frequency Domain

In this section, we investigate joint detection of UL signals from all subcarriers by extending the BSD to the space-frequency domain with the space-frequency BBCM. We prove that the space-frequency domain detector design can be converted to the that of a beam domain detector while ensuring optimality. Moreover, the asymptotically optimal space-frequency domain detector is beam structured with a low-dimensional beam domain detector for a sufficiently large

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number of BS antennas, which enables the low-complexity design and implementation.

A. Problem Formulation

Although the complexity of the proposed BSD in the space domain can be significantly reduced, it can be further simplified by jointly considering UL signals from all subcarriers and the large number of BS antennas. Note that due to the small angular spread and limited number of propagation paths of HF skywave massive MIMO-OFDM channels, the space-frequency-beam domain channel has enhanced sparsity compared to the space-beam domain channels for all subcarriers, which motivates us to extend the BSD in the space domain to the space-frequency domain to further reduce the complexity.

We consider UL transmission on the n-th OFDM symbol and omit index n in the subscript for brevity. The received signal in (3) can be rewritten as

$$\mathbf{y} = \mathbf{H}^{\mathrm{SF}}\mathbf{x} + \mathbf{z}, \tag{42}$$

where $\mathbf{y} \in \mathbb{C}^{MN_{\mathrm{v}}\times 1}$ with the $((k-k_0)M+m)$-th element $y_{m,k}$, $\mathbf{H}^{\mathrm{SF}} = \left[\mathbf{H}_0^{\mathrm{SF}}, \mathbf{H}_1^{\mathrm{SF}}, \cdots, \mathbf{H}_{U-1}^{\mathrm{SF}}\right] \in \mathbb{C}^{MN_{\mathrm{v}}\times N_{\mathrm{v}}U}$, $\mathbf{H}_u^{\mathrm{SF}} = \mathrm{vdiag}\left\{\mathbf{h}_u^{\mathrm{SF}}\right\} \in \mathbb{C}^{MN_{\mathrm{v}}\times N_{\mathrm{v}}}$, $\mathbf{x} = \left[\mathbf{x}_0^{\mathrm{T}}, \mathbf{x}_1^{\mathrm{T}}, \cdots, \mathbf{x}_{U-1}^{\mathrm{T}}\right]^{\mathrm{T}} \in \mathbb{C}^{N_{\mathrm{v}}U\times 1}$, $\mathbf{x}_u = \left[x_{u,k_0}, x_{u,k_1}, \cdots, x_{u,k_{N_{\mathrm{v}}}-1}\right]^{\mathrm{T}}$, and $\mathbf{z} \in \mathbb{C}^{MN_{\mathrm{v}}\times 1}$ is the additive white Gaussian noise with distribution $\mathcal{CN}\left(\mathbf{0}, \sigma_{\mathrm{z}}^2\mathbf{I}_{MN_{\mathrm{v}}}\right)$. The design of the conventional space-frequency domain detector $\mathbf{W}_u \in \mathbb{C}^{MN_{\mathrm{v}}\times N_{\mathrm{v}}}$ for UT u can be formulated as the following problem

$$\mathbf{W}_u^{\mathrm{op}} = \arg\min_{\mathbf{W}_u} \ \mathrm{MSE}_u, \tag{43}$$

where

$$\begin{aligned}\mathrm{MSE}_u &= \mathbb{E}\left\{\left\|\mathbf{x}_u - \mathbf{W}_u^{\mathrm{H}}\mathbf{y}\right\|^2\right\} \\ &= \mathrm{tr}\Big\{\sigma_{\mathrm{x}}^2\mathbf{I} - \sigma_{\mathrm{x}}^2\mathbf{W}_u^{\mathrm{H}}\mathbf{H}_u^{\mathrm{SF}} - \sigma_{\mathrm{x}}^2\left(\mathbf{H}_u^{\mathrm{SF}}\right)^{\mathrm{H}}\mathbf{W}_u \\ &\quad + \mathbf{W}_u^{\mathrm{H}}\Big(\sigma_{\mathrm{x}}^2\sum_{u'=0}^{U-1}\mathbf{H}_{u'}^{\mathrm{SF}}\left(\mathbf{H}_{u'}^{\mathrm{SF}}\right)^{\mathrm{H}} + \sigma_{\mathrm{z}}^2\mathbf{I}\Big)\mathbf{W}_u\Big\}.\end{aligned} \tag{44}$$

Since the space-frequency domain channel $\mathbf{H}_u^{\mathrm{SF}}$ is a vector-valued block-diagonal matrix and the received signal at different subcarriers will not interfere with each other, the optimal space-frequency domain detector should also have the vector-valued block-diagonal structure, i.e., $\mathbf{W}_u^{\mathrm{op}} = \mathrm{diag}\left\{\mathbf{w}_{u,k_0}^{\mathrm{op}}, \cdots, \mathbf{w}_{u,k_{N_{\mathrm{v}}-1}}^{\mathrm{op}}\right\}$. Actually, the problem in (43) can be decoupled into the optimization problem at each subcarrier in (26). However, to exploit sparsity of the space-frequency-beam domain channel, we focus on the space-frequency domain detector $\mathbf{W}_u$, which consists of the space domain detectors for all subcarriers.

B. BSD in the Space-Frequency Domain

According to the space-frequency BBCM in (20), the channel matrix $\mathbf{H}_u^{\mathrm{SF}}$ can be rewritten as

$$\mathbf{H}_u^{\mathrm{SF}} = \bar{\mathbf{V}}\left(\bar{\mathbf{U}}\otimes\mathbf{I}\right)\hat{\mathbf{H}}_u^{\mathrm{SF}}\bar{\mathbf{U}}^{\mathrm{H}}, \tag{45}$$

where $\hat{\mathbf{H}}_u^{\mathrm{SF}} = \mathrm{vcirc}\left\{\hat{\mathbf{h}}_u^{\mathrm{SF}}\right\} \in \mathbb{C}^{\bar{N}_{\mathrm{an}}\bar{N}_{\mathrm{de}}\times\bar{N}_{\mathrm{de}}}$ is the space-frequency-beam domain channel matrix, $\hat{\mathbf{h}}_u^{\mathrm{SF}} = \frac{1}{\sqrt{N_{\mathrm{v}}}}\left[\left(\bar{\mathbf{h}}_u^{\mathrm{SF}}\right)^{\mathrm{T}}, \mathbf{0}\right]^{\mathrm{T}} \in \mathbb{C}^{\bar{N}_{\mathrm{an}}\bar{N}_{\mathrm{de}}\times 1}$, $\bar{\mathbf{U}} = \frac{1}{\sqrt{\bar{N}_{\mathrm{de}}}}\left[\mathbf{u}\left(\tau_0\right), \mathbf{u}\left(\tau_1\right), \cdots, \mathbf{u}\left(\tau_{\bar{N}_{\mathrm{de}}-1}\right)\right] \in \mathbb{C}^{N_{\mathrm{v}}\times\bar{N}_{\mathrm{de}}}$, and $\bar{N}_{\mathrm{de}} = F_{\mathrm{de}}N_{\mathrm{v}}$. Then we have the following theorem, proved in Appendix D.

Theorem 4: The optimal solution of (43) can be obtained as

$$\mathbf{W}_u^{\mathrm{op}} = \bar{\mathbf{V}}\left(\bar{\mathbf{V}}^{\mathrm{H}}\bar{\mathbf{V}}\right)^{\dagger}\left(\bar{\mathbf{U}}\otimes\mathbf{I}\right)\tilde{\mathbf{W}}_u^{\mathrm{op}}\bar{\mathbf{U}}^{\mathrm{H}}, \tag{46}$$

where

$$\begin{aligned}\tilde{\mathbf{W}}_u^{\mathrm{op}} = \arg\min_{\tilde{\mathbf{W}}_u} \ &\mathrm{tr}\Big\{\bar{\mathbf{U}}\Big(\sigma_{\mathrm{x}}^2\mathbf{I} - \sigma_{\mathrm{x}}^2\tilde{\mathbf{W}}_u^{\mathrm{H}}\hat{\mathbf{H}}_u^{\mathrm{SF}} - \sigma_{\mathrm{x}}^2\Big(\hat{\mathbf{H}}_u^{\mathrm{SF}}\Big)^{\mathrm{H}}\tilde{\mathbf{W}}_u \\ &+ \sigma_{\mathrm{x}}^2\sum_{u'=0}^{U-1}\tilde{\mathbf{W}}_u^{\mathrm{H}}\hat{\mathbf{H}}_{u'}^{\mathrm{SF}}\Big(\hat{\mathbf{H}}_{u'}^{\mathrm{SF}}\Big)^{\mathrm{H}}\tilde{\mathbf{W}}_u \\ &+ \sigma_{\mathrm{z}}^2\tilde{\mathbf{W}}_u^{\mathrm{H}}\left(\bar{\mathbf{U}}^{\mathrm{H}}\otimes\mathbf{I}\right)\left(\bar{\mathbf{V}}^{\mathrm{H}}\bar{\mathbf{V}}\right)^{\dagger}\left(\bar{\mathbf{U}}\otimes\mathbf{I}\right)\tilde{\mathbf{W}}_u\Big)\bar{\mathbf{U}}^{\mathrm{H}}\Big\} \\ \text{s.t. } &\bar{\mathbf{V}}^{\mathrm{H}}\left(\bar{\mathbf{V}}^{\mathrm{H}}\right)^{\dagger}\left(\bar{\mathbf{U}}\otimes\mathbf{I}\right)\tilde{\mathbf{W}}_u\bar{\mathbf{U}}^{\mathrm{H}} = \left(\bar{\mathbf{U}}\otimes\mathbf{I}\right)\tilde{\mathbf{W}}_u\bar{\mathbf{U}}^{\mathrm{H}}.\end{aligned} \tag{47}$$

Theorem 4 provides an optimal detector structure. In the optimization problem in (47), the space-frequency-beam domain channel matrix $\hat{\mathbf{H}}_u^{\mathrm{SF}}$ instead of the space-frequency domain channel matrix $\mathbf{H}_u^{\mathrm{SF}}$ is involved and we can turn to optimizing $\tilde{\mathbf{W}}_u$ instead of $\mathbf{W}_u$ to minimize the MSE. When we obtain $\tilde{\mathbf{W}}_u^{\mathrm{op}}$ by solving the optimization problem in (47), we can directly construct detector $\mathbf{W}_u^{\mathrm{op}}$ according to the relation in (46). Moreover, since $\lim_{M\to\infty}\bar{\mathbf{V}}_k^{\mathrm{H}}\left(\bar{\mathbf{V}}_k^{\mathrm{H}}\right)^{\dagger} - \mathbf{I} = \mathbf{0}$ and $\lim_{M\to\infty}\left(\bar{\mathbf{V}}_k^{\mathrm{H}}\bar{\mathbf{V}}_k\right)^{\dagger} = \mathbf{I}$, when $M\to\infty$, the optimal solution in (46) asymptotically becomes

$$\mathbf{W}_u^{\mathrm{op},\infty} = \bar{\mathbf{V}}\left(\bar{\mathbf{U}}\otimes\mathbf{I}\right)\tilde{\mathbf{W}}_u^{\mathrm{op},\infty}\bar{\mathbf{U}}^{\mathrm{H}}, \tag{48}$$

where

$$\begin{aligned}&\tilde{\mathbf{W}}_u^{\mathrm{op},\infty} \\ &= \arg\min_{\tilde{\mathbf{W}}_u}\mathrm{tr}\Big\{\bar{\mathbf{U}}\Big(\sigma_{\mathrm{x}}^2\mathbf{I} - \sigma_{\mathrm{x}}^2\tilde{\mathbf{W}}_u^{\mathrm{H}}\hat{\mathbf{H}}_u^{\mathrm{SF}} - \sigma_{\mathrm{x}}^2\Big(\hat{\mathbf{H}}_u^{\mathrm{SF}}\Big)^{\mathrm{H}}\tilde{\mathbf{W}}_u \\ &\quad + \sigma_{\mathrm{x}}^2\tilde{\mathbf{W}}_u^{\mathrm{H}}\Big(\sum_{u'=0}^{U-1}\hat{\mathbf{H}}_{u'}^{\mathrm{SF}}\Big(\hat{\mathbf{H}}_{u'}^{\mathrm{SF}}\Big)^{\mathrm{H}} + \sigma_{\mathrm{z}}^2\mathbf{I}\Big)\tilde{\mathbf{W}}_u\Big)\bar{\mathbf{U}}^{\mathrm{H}}\Big\}.\end{aligned} \tag{49}$$

The relation in (48) provides an asymptotically optimal detector structure for $\mathbf{W}_u^{\mathrm{op},\infty}$. Since only space-frequency-beam domain channels are involved in (49) and the structure in (48) is very similar to (45), which is established according to the space-frequency BBCM, we refer to $\tilde{\mathbf{W}}_u^{\mathrm{op},\infty}$ also as the beam domain detector, and the resulting space-frequency domain detector, $\mathbf{W}_u^{\mathrm{op},\infty}$, is referred to as the BSD in the space-frequency domain.

Although the optimization variable $\tilde{\mathbf{W}}_u$ is a vector-circulant matrix like the beam domain channel matrix $\hat{\mathbf{H}}_u^{\mathrm{SF}}$, all $\bar{N}_{\mathrm{an}}\bar{N}_{\mathrm{de}}$ beams should be considered and the complexity of the search for $\tilde{\mathbf{W}}_u^{\mathrm{op},\infty}$ is still very high. Therefore, we have the following Theorem, proved in Appendix E, which can significantly reduce the computational complexity of $\tilde{\mathbf{W}}_u^{\mathrm{op},\infty}$ under certain conditions.

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Theorem 5: When $\forall u, n_{\rm an}, \left|\mathcal{N}^{\rm de}_{u,n_{\rm an}}\right| \leq 1$ and $\forall u, u', \mathcal{N}^{\rm an}_{u} \cap \mathcal{N}^{\rm an}_{u'} = \emptyset$, the optimal solution of (49) can be obtained as

$$\tilde{\mathbf{W}}^{\rm op,\infty}_{u} = \left(\left\|\hat{\mathbf{h}}^{\rm SF}_{u}\right\|^2 + \frac{\sigma^2_{\rm z}}{\sigma^2_{\rm x}}\right)^{-1} \hat{\mathbf{H}}^{\rm SF}_{u}. \tag{50}$$

From Theorem 5, if there are no two paths with the same directional cosine but different delays for each user and there are no overlapping beam domain channels for any two UTs, the optimal beam domain detector is the scaled beam domain channel matrix $\hat{\mathbf{H}}^{\rm SF}_{u}$, while the scaling factor is related to the energy of the corresponding space-frequency-beam domain channel vector. According to (48), the space-frequency domain detector is

$$\mathbf{W}^{\rm op,\infty}_{u} = \left(\left\|\hat{\mathbf{h}}^{\rm SF}_{u}\right\|^2 + \frac{\sigma^2_{\rm z}}{\sigma^2_{\rm x}}\right)^{-1} \bar{\mathbf{V}} \left(\bar{\mathbf{U}} \otimes \mathbf{I}\right) \hat{\mathbf{H}}^{\rm SF}_{u} \bar{\mathbf{U}}^{\rm H}, \tag{51}$$

and the detection result is presented as

$$\begin{aligned}\hat{\mathbf{x}}_u &= \left(\mathbf{W}^{\rm op,\infty}_{u}\right)^{\rm H}\mathbf{y} \\ &= \left(\left\|\hat{\mathbf{h}}^{\rm SF}_{u}\right\|^2 + \frac{\sigma^2_{\rm z}}{\sigma^2_{\rm x}}\right)^{-1} \bar{\mathbf{U}}\left(\hat{\mathbf{H}}^{\rm SF}_{u}\right)^{\rm H} \left(\bar{\mathbf{U}}^{\rm H} \otimes \mathbf{I}\right) \bar{\mathbf{V}}^{\rm H}\mathbf{y}.\end{aligned} \tag{52}$$

According to the BBCMs (19) and (20), we have

$$\left(\bar{\mathbf{U}} \otimes \mathbf{I}\right) \hat{\mathbf{H}}^{\rm SF}_{u} \bar{\mathbf{U}}^{\rm H} = {\rm diag}\left\{\bar{\mathbf{h}}^{\rm S}_{u,0}, \cdots, \bar{\mathbf{h}}^{\rm S}_{u,N_{\rm v}-1}\right\}. \tag{53}$$

Substituting (53) into (52), we have

$$\hat{\mathbf{x}}_u = \left(\left\|\hat{\mathbf{h}}^{\rm SF}_{u}\right\|^2 + \frac{\sigma^2_{\rm z}}{\sigma^2_{\rm x}}\right)^{-1} {\rm diag}\left\{\bar{\mathbf{h}}^{\rm S}_{u,0}, \cdots, \bar{\mathbf{h}}^{\rm S}_{u,N_{\rm v}-1}\right\}^{\rm H} \bar{\mathbf{V}}^{\rm H}\mathbf{y}. \tag{54}$$

C. Complexity Analysis

In this subsection, we analyze the complexity of the proposed BSD in the space-frequency domain. We define the average value of $\bar{N}_{{\rm an},u}$ and $\bar{N}_{{\rm SFB},u}$ as $\bar{N}^{\rm ave}_{\rm an} = \frac{1}{U}\sum_{u=0}^{U-1} \bar{N}_{{\rm an},u}$ and $\bar{N}^{\rm ave}_{\rm SFB} = \frac{1}{U}\sum_{u=0}^{U-1} \bar{N}_{{\rm SFB},u}$ for brecity. We first analyze the design complexity of the detector, which comes from (50). Since the optimal beam domain detector is the scaled beam domain channel matrix, all we need is to calculate the scaling factor $\left(\left\|\hat{\mathbf{h}}^{\rm SF}_{u}\right\|^2 + \frac{\sigma^2_{\rm z}}{\sigma^2_{\rm x}}\right)^{-1}$. Consequently the design complexity is $\mathcal{O}\left(\bar{N}^{\rm ave}_{\rm SFB} U\right)$, which is very low thanks to the sparsity of the space-frequency-beam domain channel for HF skywave massive MIMO-OFDM channels [7]. As a comparison, the direct design of space-frequency domain detector is typically performed by designing the space domain MMSE detector in (23) at each subcarrier, whose design complexity is $\mathcal{O}\left(\left(MU^2 + U^3\right) N_{\rm v}\right)$.

The implementation complexity of the proposed BSD in the space-frequency domain comes from (54). It can be implemented by calculating $\left(\left\|\hat{\mathbf{h}}^{\rm SF}_{u}\right\|^2 + \frac{\sigma^2_{\rm z}}{\sigma^2_{\rm x}}\right)^{-1} \left(\bar{\mathbf{h}}^{\rm S}_{u,k}\right)^{\rm H} \bar{\mathbf{V}}^{\rm H}_{k} \mathbf{y}_k$ at each subcarrier. Similar to the BSD in the space domain, the computation of $\bar{\mathbf{V}}^{\rm H}_{k}\mathbf{y}_k$ can be implemented with CZT. Thus the implementation complexity of the BSD in the space-frequency domain is $\mathcal{O}\left(\left(N_{\rm (S)}\log_2 N_{\rm (S)} + \bar{N}^{\rm ave}_{\rm an} U\right) N_{\rm v}\right)$ while the implementation complexity of the MMSE detector is $O\left(MUN_{\rm v}\right)$.

It should be noted that although the complexity of the proposed BSD in the space-frequency domain is extremely low due to the utilization of the channel sparsity in the space-frequency-beam domain, it works well only in scenarios where the conditions in Theorem 5 are satisfied. In other words, the space-frequency-beam domain channels of the served UTs should be very sparse and the served UTs should be scheduled to make the beam domain channels non-overlapping. If these conditions are not satisfied, such as in scenarios where the number of UTs is large, the BSD in the space-frequency domain may not work well. In this case, we can turn to using the BSD in the space domain, which allows the beam domain channels of different UTs to overlap and where more beams can be selected according to the criterion in Theorem 3 to improve the detection performance.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Carrier frequency $f_{\rm c}$	16 MHz
Subcarrier spacing Δ_f	250 Hz
Number of subcarriers $N_{\rm c}$	2048
Length of CP $N_{\rm g}$	512
Number of BS antennas M	128 / 256 / 512
BS antenna spacing d	9 m
Number of subcarriers in use $N_{\rm v}$	1536
Modulation scheme	16-QAM

V. SIMULATION RESULTS AND DISCUSSIONS

In this section, we provide simulation results to illustrate the efficiency of the proposed detectors for HF skywave massive MIMO-OFDM communications.

A. Simulation Setup

The main simulation parameters are given in Table I. We generate UTs at a distance of 2,000 km from the BS, and azimuth angles of the UTs seen from the BS are generated in the interval $(-90^\circ\ 90^\circ)$. The realistic HF skywave channel parameters are generated for each UT independently by a commercial ray-tracing software Proplab-Pro version 3.1 [44]. Specifically, the software is utilized to generate parameters of each propagation path between the BS and each UT, including the path gain, the propagation delay, the azimuth and elevation AoA. The maximum Doppler is $\nu_{\rm max} = \nu_{\rm iono}/2 + v f_{\rm c}/c$, where $\nu_{\rm iono}$ is the Doppler spread imposed by the ionosphere and set to be 0.5 Hz [45] while v is the velocity of UTs and set to be 100 km/h. The Doppler frequencies for paths are generated within $[-\nu_{\rm max}, \nu_{\rm max})$. With these parameters, it can be verified that the duration of an OFDM symbol, $T_{\rm sym} = 0.005$ s, is much less than the channel coherence time $T_{\rm c} = 1/(2\nu_{\rm max}) \approx 0.289$ s, which justifies the assumption that the channel keeps constant within an OFDM symbol. The initial phases for paths are generated randomly. Then using the generated parameters for paths, the beam based channels in the space domain and in the space-frequency domain can be generated according to (11) and (13), respectively, where fine factors $F_{\rm an}$ and $F_{\rm de}$ are set to two [7]. Moreover, using a similar approach in [7], we schedule the UTs to select those with relatively low channel correlation for simulation.

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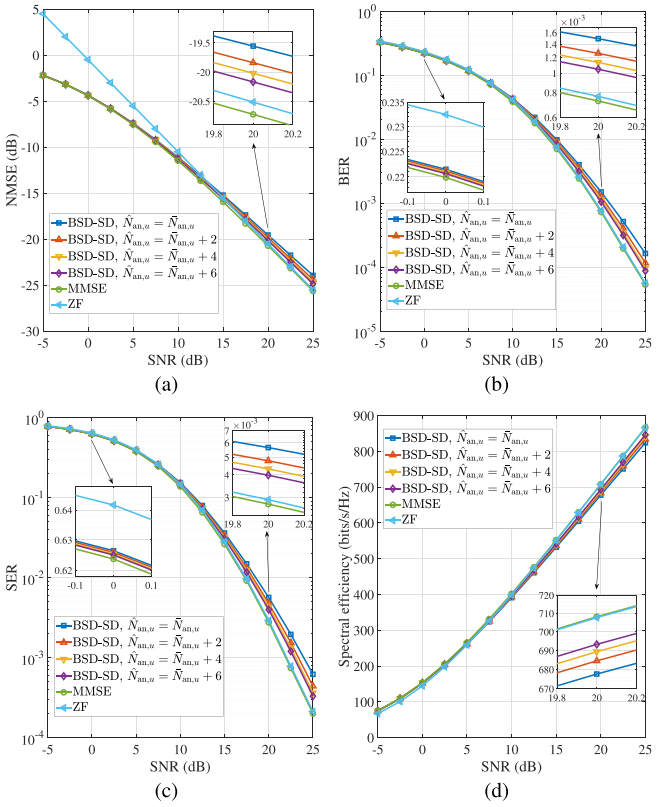


Fig. 3. The performance of BSD-SD with different numbers of selected beams, MMSE and ZF. (a) NMSE; (b) BER; (c) SER; (d) Spectral efficiency.

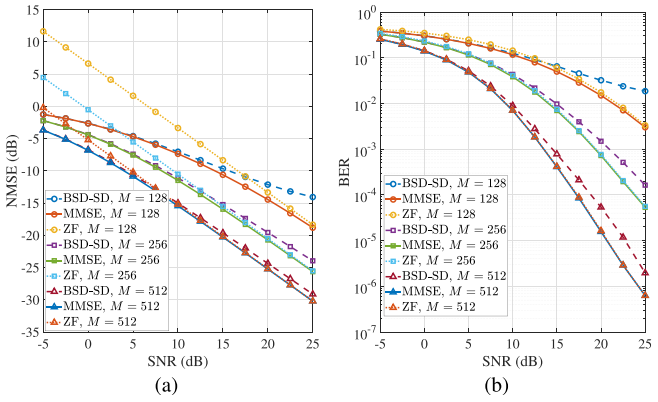


Fig. 4. The performance of BSD-SD, MMSE and ZF with different numbers of BS antennas. (a) NMSE; (b) BER.

The signal-to-noise ratio (SNR) in the simulation refers to the received SNR. The modulation scheme is 16 quadrature amplitude modulation (QAM). For brevity, BSD in the space domain and in the space-frequency domain are referred to as BSD-SD and BSD-SFD, respectively. The MMSE detector and the ZF detector [21] are included for comparison.

B. BSD-SD

Fig. 3 shows the normalized mean-squared error (NMSE), bit error rate (BER), data symbol error rate (SER) and spectral efficiency performance of the BSD-SD with different numbers of selected beams, where $M = 256$ and $U = 96$. Note that

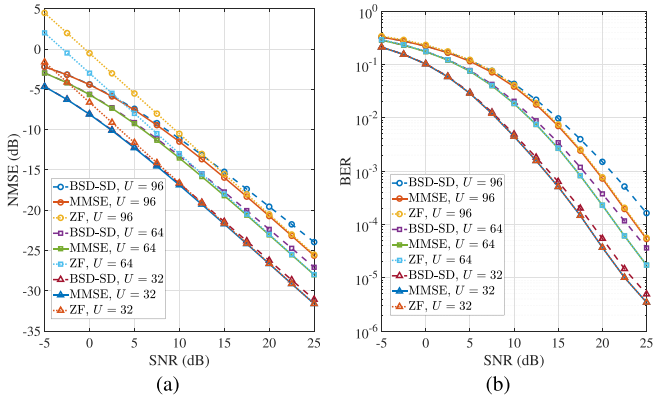


Fig. 5. The performance of BSD-SD, MMSE and ZF with different numbers of UTs. (a) NMSE; (b) BER.

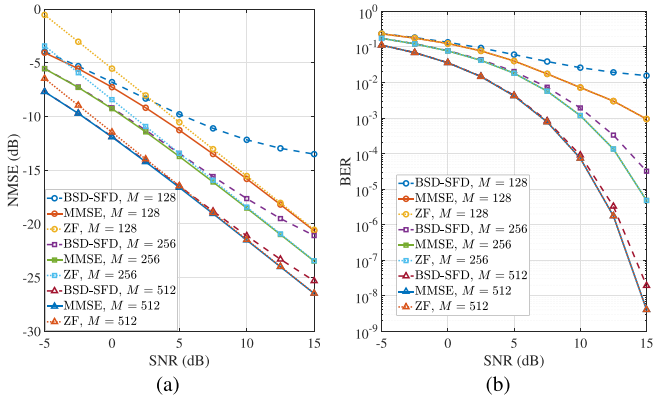


Fig. 6. The performance of BSD-SFD, MMSE and ZF with different numbers of BS antennas. (a) NMSE; (b) BER.

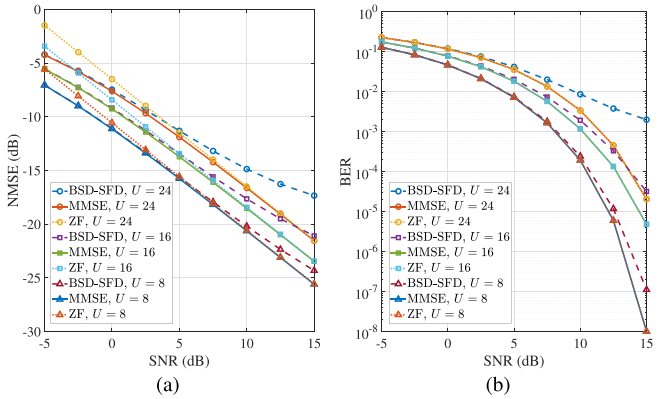


Fig. 7. The performance of BSD-SFD, MMSE and ZF detector with different numbers of UTs. (a) NMSE; (b) BER.

the spectral efficiency is calculated by [6]

$$\mathcal{R} = \frac{1}{N_{\mathrm{v}}} \sum_{k \in \mathcal{K}} \sum_{u=0}^{U-1} \mathbb{E}\left\{ \log_2 \left(1 + \frac{\left| \mathbf{w}_{u,k}^{\mathrm{H}} \mathbf{h}_{u,k}^{\mathrm{S}} \right|^2 \sigma_{\mathrm{x}}^2}{\sum_{u'=0, u' \neq u}^{U-1} \left| \mathbf{w}_{u,k}^{\mathrm{H}} \mathbf{h}_{u',k}^{\mathrm{S}} \right|^2 \sigma_{\mathrm{x}}^2 + \sigma_{\mathrm{z}}^2 \| \mathbf{w}_{u,k} \|^2} \right) \right\}, \quad (55)$$

where $\mathbf{w}_{u,k}$ can be the MMSE detector, the ZF detector or the BSD-SD. We select beams for the BSD-SD according to Theorem 3. From the figure, the performance of the proposed

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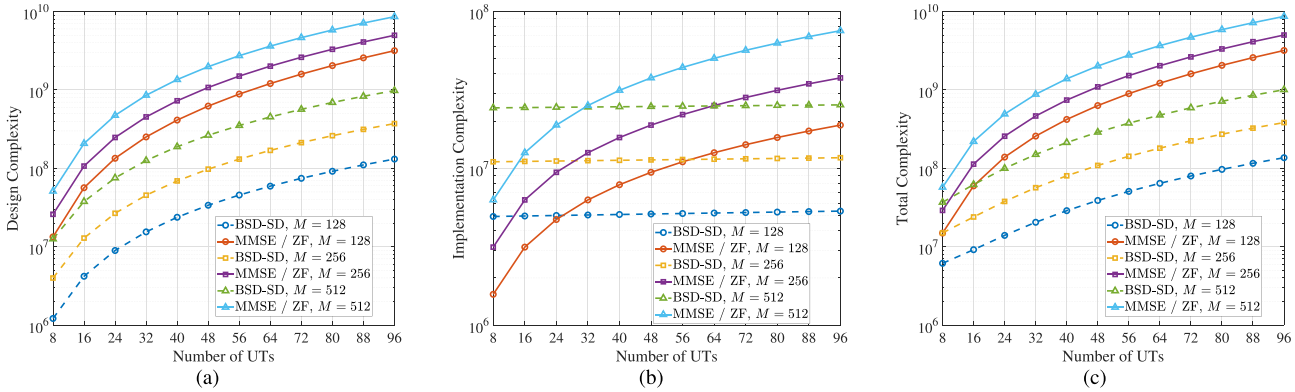


Fig. 8. The complexities of BSD-SD, MMSE and ZF versus the number of UTs. (a) Design complexity; (b) Implementation complexity; (c) Total complexity.

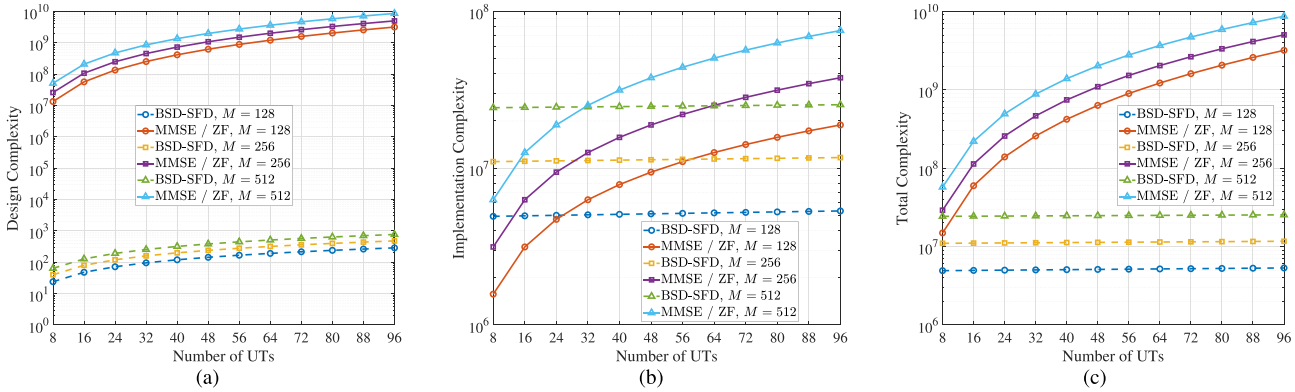


Fig. 9. The complexities of BSD-SFD, MMSE and ZF versus the number of UTs. (a) Design complexity; (b) Implementation complexity; (c) Total complexity.

BSD-SD approaches that of the MMSE and ZF detectors and the performance gap increases with SNR due to the inter-UT interference. However, the performance gap between the BSD-SD and the MMSE or ZF detector decreases as the number of selected beams increases. According to Theorem 3, when there exists overlap between the beam domain channels of different UTs, we use more beams to improve the performance of the BSD-SD at the cost of higher complexity, thus reducing the performance gap.

Fig. 4 shows the performance of the BSD-SD, the MMSE and ZF detectors with different numbers of BS antennas, where $U = 96$ and $\hat{N}_{\mathrm{an},u} = \bar{N}_{\mathrm{an},u}$. From the figure, the performance gap between the BSD-SD and the MMSE or ZF detector increases with SNR since the impact of the inter-UT interference becomes obvious in the high SNR region, especially when the number of BS antennas is $M = 128$ at which the deterioration of the orthogonality between the beams results in a larger inter-UT interference. However, the performance gap decreases as the number of BS antennas increases. This is because when M becomes large, the beams become approximately orthogonal, which mitigates inter-UT interference and thus reduces the performance gap. According to Theorem 2, it can be expected that as M becomes sufficiently large, the performance gap will tend to disappear if UTs can be well scheduled to make the beam domain channels non-overlapping.

Fig. 5 shows the performance of the BSD-SD, the MMSE and ZF detectors with different numbers of UTs, where $M = 256$ and $\hat{N}_{\mathrm{an},u} = \bar{N}_{\mathrm{an},u}$. From the figure, the performance gap between the BSD-SD and the MMSE or ZF detector decreases as the number of UTs decreases. This is because when decreasing the number of UTs, the UTs can be more easily scheduled to reduce the beam domain channel overlap degree, which can reduce inter-UT interference and result in a lower performance gap. Besides, the BSD-SD still has a satisfactory performance, approaching that of the MMSE and ZF detectors even when the number of UTs is large.

C. BSD-SFD

Fig. 6 shows the performance of the BSD-SFD, the MMSE and ZF detectors with different numbers of BS antennas, where $U = 16$. Similar to the BSD-SD, the performance gap between the BSD-SFD and the MMSE or ZF detector increases with SNR due to the inter-UT interference, particularly when M is reduced. However, such a performance gap decreases as the number of BS antennas increases. This is because when M becomes large, the columns of $\bar{\mathbf{V}}_k$ can be nearly orthogonal and the approximation in (48) can be more accurate. Moreover, such an asymptotic orthogonality can also reduce the inter-UT interference and allow the condition in Theorem 5 to be better satisfied and improve the performance of the BSD-SFD.

Fig. 7 shows the performance of the BSD-SFD, the MMSE and ZF detectors with different numbers of UTs, where $M = 256$. From the figure, the performance gap between the BSD-SFD and the MMSE or ZF detector decreases as the

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number of UTs decreases. This is because fewer number of UTs reduce the inter-UT interference and make the condition in Theorem 5 better satisfied, which leads to a lower performance gap.

D. Complexity

Fig. 8 and Fig. 9 show the complexity of different detectors, where $\hat{N}_{\mathrm{an},u} = \bar{N}_{\mathrm{an},u}$. The complexities in Fig. 8 and Fig. 9 refer to the computational complexity for all subcarriers. The total complexity refers to the sum of the design complexity and the implementation complexity. Note that the ZF detector has the same design and implementation complexities as the MMSE detector [21]. From the figures, both the BSD-SD and BSD-SFD have lower complexity than the MMSE and ZF detectors due to the utilization of the channel sparsity in the space-beam domain and space-frequency-beam domain. Moreover, the BSD-SFD achieves the lowest complexity since it considers two-dimensional (space-frequency-beam domain) sparsity instead of one-dimensional (space-beam domain) sparsity considered in the BSD-SD. However, it is noted that the served number of UTs with the BSD-SFD is fewer than that with the BSD-SD due to the more stricter conditions in Theorem 5. Therefore, when the number of UTs is small and the space-frequency-beam domain channel has a significant sparsity, we can choose the BSD-SFD, which has a good performance with much lower complexity. Otherwise, when the number of UTs becomes large, we can turn to the BSD-SD, whose complexity is also much lower than that of the MMSE and ZF detectors while ensuring a satisfactory performance.

VI. Conclusion

In this paper, we have investigated signal detection for HF skywave massive MIMO-OFDM communications. We first introduced the BBCMs and revealed the sparsity of the space-beam domain channel and the space-frequency-beam domain channel, respectively. Based on the space BBCM, we proposed a BSD in the space domain for each subcarrier, where we proved that the space domain detector can be designed by optimizing a low-dimensional beam domain detector, and corresponding beam selection criterion was also provided. Moreover, based on the space-frequency BBCM, we extended the BSD to the joint space-frequency domain for all subcarriers, where the space-frequency domain detector design can also be converted to that of a low-dimensional beam domain detector. Furthermore, both the proposed detectors can be efficiently implemented by exploiting the beam structure. Simulation results verified the effectiveness of proposed detectors.

Appendix A
Proof of Theorem 1

Define $\tilde{\mathbf{w}}_{u,k} = \bar{\mathbf{V}}_k^{\mathrm{H}} \mathbf{w}_{u,k} \in \mathbb{C}^{\bar{N}_{\mathrm{an}} \times 1}$, (27) is rewritten as

$$\begin{aligned}\mathrm{MSE}_{u,k} = {} & \sigma_{\mathrm{x}}^2 - \sigma_{\mathrm{x}}^2 \tilde{\mathbf{w}}_{u,k}^{\mathrm{H}} \bar{\mathbf{h}}_{u,k}^{\mathrm{S}} - \sigma_{\mathrm{x}}^2 \big(\bar{\mathbf{h}}_{u,k}^{\mathrm{S}}\big)^{\mathrm{H}} \tilde{\mathbf{w}}_{u,k} \\ & + \sigma_{\mathrm{x}}^2 \sum_{u'=0}^{U-1} \tilde{\mathbf{w}}_{u,k}^{\mathrm{H}} \bar{\mathbf{h}}_{u',k}^{\mathrm{S}} \big(\bar{\mathbf{h}}_{u',k}^{\mathrm{S}}\big)^{\mathrm{H}} \tilde{\mathbf{w}}_{u,k} + \sigma_{\mathrm{z}}^2 \mathbf{w}_{u,k}^{\mathrm{H}} \mathbf{w}_{u,k}. \end{aligned} \tag{56}$$

In (56), $\mathbf{w}_{u,k}$ in most terms is replaced by $\tilde{\mathbf{w}}_{u,k}$. Although $\mathbf{w}_{u,k}$ is still involved in the term $\sigma_{\mathrm{z}}^2 \mathbf{w}_{u,k}^{\mathrm{H}} \mathbf{w}_{u,k}$, it can be obtained according to the relation $\tilde{\mathbf{w}}_{u,k} = \bar{\mathbf{V}}_k^{\mathrm{H}} \mathbf{w}_{u,k}$ when $\tilde{\mathbf{w}}_{u,k}$ is given. Therefore, to ensure the existence of $\mathbf{w}_{u,k}$ such that $\tilde{\mathbf{w}}_{u,k} = \mathbf{V}_k^{\mathrm{H}} \mathbf{w}_{u,k}$ when $\tilde{\mathbf{w}}_{u,k}$ is given, $\tilde{\mathbf{w}}_{u,k}$ should satisfy the constraint $\big(\bar{\mathbf{V}}_k^H \big(\bar{\mathbf{V}}_k^H\big)^\dagger - \mathbf{I}\big)\tilde{\mathbf{w}}_{u,k} = \mathbf{0}$ [46]. Therefore, we can turn to optimizing $\tilde{\mathbf{w}}_{u,k}$ instead of $\mathbf{w}_{u,k}$ to minimize the MSE. Moreover, note that for a given optimal solution $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op}}$, the corresponding optimal detector $\mathbf{w}_{u,k}^{\mathrm{op}}$ should have the mimimal power, i.e., $\big(\mathbf{w}_{u,k}^{\mathrm{op}}\big)^{\mathrm{H}} \mathbf{w}_{u,k}^{\mathrm{op}}$ should be minimal, and we can obtain the corresponding $\mathrm{MSE}_{u,k}$ in (56). Otherwise, we will have a $\mathbf{w}_{u,k}^{\mathrm{op}}$ with lower power, which leads to a lower $\mathrm{MSE}_{u,k}$ and contradicts the optimality. Therefore, we could formulate the following optimization problem

$$\mathbf{w}_{u,k}^{\mathrm{op}} = \arg\min_{\mathbf{w}_{u,k}} \ \mathbf{w}_{u,k}^{\mathrm{H}} \mathbf{w}_{u,k}, \quad \text{s.t. } \mathbf{V}_k^{\mathrm{H}} \mathbf{w}_{u,k} = \tilde{\mathbf{w}}_{u,k}^{\mathrm{op}}. \tag{57}$$

The optimal solution of problem (57) can be obtained as

$$\mathbf{w}_{u,k}^{\mathrm{op}} = \bar{\mathbf{V}}_k \big(\bar{\mathbf{V}}_k^{\mathrm{H}} \bar{\mathbf{V}}_k\big)^\dagger \tilde{\mathbf{w}}_{u,k}^{\mathrm{op}}. \tag{58}$$

Then according to (58), we replace $\mathbf{w}_{u,k}$ with $\bar{\mathbf{V}}_k \big(\bar{\mathbf{V}}_k^{\mathrm{H}} \bar{\mathbf{V}}_k\big)^\dagger \tilde{\mathbf{w}}_{u,k}$ in (56) and obtain the objective function in (29). This completes the proof.

Appendix B
Proof of Theorem 2

Define $\mathbf{s}_u \in \mathbb{C}^{\bar{N}_{\mathrm{an}} \times 1}$ as the beam selection vector with

$$[\mathbf{s}_u]_{n_{\mathrm{an}}} = \begin{cases} 1, & [\mathcal{N}^{\mathrm{an}}]_{n_{\mathrm{an}}} \in \mathcal{N}_u^{\mathrm{an}} \\ 0, & \text{otherwise.} \end{cases} \tag{59}$$

We denote $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}$ as the optimal solution and define $\bar{\mathbf{w}}_{u,k} = \tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty} \circ \mathbf{s}_u$, thus we have

$$\mathrm{MSE}_{u,k}\big(\bar{\mathbf{w}}_{u,k}\big) \geq \mathrm{MSE}_{u,k}\big(\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}\big). \tag{60}$$

On the other hand, we have $\sum_{u'=0}^{U-1} \bar{\mathbf{w}}_{u,k}^{\mathrm{H}} \bar{\mathbf{h}}_{u',k}^{\mathrm{S}} \big(\bar{\mathbf{h}}_{u',k}^{\mathrm{S}}\big)^{\mathrm{H}} \bar{\mathbf{w}}_{u,k} \leq \sum_{u'=0}^{U-1} \big(\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}\big)^{\mathrm{H}} \bar{\mathbf{h}}_{u',k}^{\mathrm{S}} \big(\bar{\mathbf{h}}_{u',k}^{\mathrm{S}}\big)^{\mathrm{H}} \tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}$, $\bar{\mathbf{w}}_{u,k}^{\mathrm{H}} \bar{\mathbf{w}}_{u,k} \leq \big(\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}\big)^{\mathrm{H}} \tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}$, and $\bar{\mathbf{w}}_{u,k}^{\mathrm{H}} \bar{\mathbf{h}}_{u,k}^{\mathrm{S}} = \big(\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}\big)^{\mathrm{H}} \bar{\mathbf{h}}_{u,k}^{\mathrm{S}}$, which leads to

$$\mathrm{MSE}_{u,k}\big(\bar{\mathbf{w}}_{u,k}\big) \leq \mathrm{MSE}_{u,k}\big(\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}\big). \tag{61}$$

Combining (60) and (61), we have $\mathrm{MSE}_{u,k}\big(\bar{\mathbf{w}}_{u,k}\big) = \mathrm{MSE}_{u,k}\big(\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}\big)$, which means that $\tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty} = \bar{\mathbf{w}}_{u,k} = \tilde{\mathbf{w}}_{u,k}^{\mathrm{op},\infty} \circ \mathbf{s}_u$ and we can obtain (32). This completes the proof.

Appendix C
Proof of Theorem 3

When $M \to \infty$, since $\bar{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}$ is the MMSE solution, we can easily obtain that

$$\begin{aligned}&\mathrm{MSE}_{u,k}\big(\mathbf{w}_{u,k}^{\mathrm{op},\infty}\big(\mathcal{N}_{u,1}^{\mathrm{an}}\big)\big) \\ &= \sigma_{\mathrm{x}}^2 \Big(1 - \big[\tilde{\mathbf{h}}_k^{\mathrm{S}}\big]_{\mathcal{N}_{u,1}^{\mathrm{an}}}^{\mathrm{H}} \Big(\big[\tilde{\mathbf{H}}_k^{\mathrm{S}}\big]_{\mathcal{N}_{u,1}^{\mathrm{an}},:} \big[\tilde{\mathbf{H}}_k^{\mathrm{S}}\big]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}^{\mathrm{H}} + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2} \mathbf{I} \Big)^{-1} \\ &\quad \times \big[\tilde{\mathbf{h}}_k^{\mathrm{S}}\big]_{\mathcal{N}_{u,1}^{\mathrm{an}}} \Big), \end{aligned} \tag{62}$$

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$$\begin{aligned}&\mathrm{MSE}_{u,k}\left(\mathbf{w}_{u,k}^{\mathrm{op},\infty}\left(\mathcal{N}_{u,2}^{\mathrm{an}}\right)\right)\\&=\sigma_{\mathrm{x}}^2\Big(1-\left(\tilde{\mathbf{h}}_{u,k,\tilde{n}_{\mathrm{an}}}^{\mathrm{S}}\right)^{\mathrm{H}}\Big(\tilde{\mathbf{H}}_{u,k,\tilde{n}_{\mathrm{an}}}^{\mathrm{S}}\left(\tilde{\mathbf{H}}_{u,k,\tilde{n}_{\mathrm{an}}}^{\mathrm{S}}\right)^{\mathrm{H}}+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\Big)^{-1}\\&\quad\times\tilde{\mathbf{h}}_{u,k,\tilde{n}_{\mathrm{an}}}^{\mathrm{S}}\Big),\end{aligned}\tag{63}$$

where $\tilde{\mathbf{h}}_{u,k,\tilde{n}_{\mathrm{an}}}^{\mathrm{S}}=\left[0,\left[\tilde{\mathbf{h}}_{u,k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}}}^{\mathrm{T}}\right]^{\mathrm{T}}$, $\tilde{\mathbf{H}}_{k}^{\mathrm{S}}=\left[\tilde{\mathbf{h}}_{0,k}^{\mathrm{S}},\cdots,\tilde{\mathbf{h}}_{U-1,k}^{\mathrm{S}}\right]$ and $\tilde{\mathbf{H}}_{k,\tilde{n}_{\mathrm{an}}}^{\mathrm{S}}=\left[\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}^{\mathrm{T}},\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}^{\mathrm{T}}\right]^{\mathrm{T}}$. We have (64), shown in the equation at the bottom of the page. Thus according to [47, Prop. 14.11], we have in (64), shown at the bottom of the page, where $\mathbf{A}_1$, $\mathbf{A}_2$ and $\mathbf{A}_3$ are the matrices with proper dimensions. Then (63) can be rewritten as

$$\begin{aligned}\sigma_{\mathrm{x}}^2\Bigg(1-\left[\tilde{\mathbf{h}}_{u,k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}}}^{\mathrm{H}}\Bigg(\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}\Big(\frac{\sigma_{\mathrm{x}}^2}{\sigma_{\mathrm{z}}^2}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}^{\mathrm{H}}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}+\mathbf{I}\Big)^{-1}\\\times\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}^{\mathrm{H}}+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\Big)^{-1}\left[\tilde{\mathbf{h}}_{u,k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}}}\Bigg).\end{aligned}$$

Note that

$$\begin{aligned}&\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}^{\mathrm{H}}+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\\&\succeq\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}\Big(\frac{\sigma_{\mathrm{x}}^2}{\sigma_{\mathrm{z}}^2}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}^{\mathrm{H}}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}+\mathbf{I}\Big)^{-1}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}^{\mathrm{H}}\\&\quad+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}.\end{aligned}$$

Then, according to [47, Prop. 10.51], we have

$$\begin{aligned}&\Big(\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}^{\mathrm{H}}+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\Big)^{-1}\\&\preceq\Big(\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}\Big(\frac{\sigma_{\mathrm{x}}^2}{\sigma_{\mathrm{z}}^2}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}^{\mathrm{H}}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}+\mathbf{I}\Big)^{-1}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}^{\mathrm{H}}\\&\quad+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\Big)^{-1}.\end{aligned}$$

Thus we have $\mathrm{MSE}_{u,k}\left(\bar{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}\left(\mathcal{N}_{u,1}^{\mathrm{an}}\right)\right)\geq\mathrm{MSE}_{u,k}\left(\bar{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}\left(\mathcal{N}_{u,2}^{\mathrm{an}}\right)\right)$.

When $\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}^{\mathrm{H}}=\mathbf{0}$, we have (66), shown in the equation at the bottom of the page. Then we have $\mathrm{MSE}_{u,k}\left(\bar{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}\left(\mathcal{N}_{u,1}^{\mathrm{an}}\right)\right)=\mathrm{MSE}_{u,k}\left(\bar{\mathbf{w}}_{u,k}^{\mathrm{op},\infty}\left(\mathcal{N}_{u,2}^{\mathrm{an}}\right)\right)$. This completes the proof.

Appendix D
Proof of Theorem 4

Before proceeding, we provide the following two results that are useful in the subsequent discussion.

Lemma 1 [46, Thm. 2]: For any $\mathbf{A}\in\mathbb{C}^{N_1\times N_2}$, the sufficient and necessary condition for the existence of $\mathbf{B}\in\mathbb{C}^{\tilde{N}_1\times\tilde{N}_2}$ such that $\mathbf{A}=\mathbf{CBD}$ is

$$\mathbf{CC}^{\dagger}\mathbf{AD}^{\dagger}\mathbf{D}=\mathbf{A}.\tag{65}$$

Lemma 2: If $\mathbf{C}_1,\mathbf{C}_2\in\mathbb{C}^{\bar{N}_{\mathrm{an}}\bar{N}_{\mathrm{de}}\times\bar{N}_{\mathrm{de}}}$ are vector-valued block-circulant matrices, we have

$$\bar{\mathbf{U}}\mathbf{C}_1^{\mathrm{H}}\left(\bar{\mathbf{U}}^{\mathrm{H}}\bar{\mathbf{U}}\otimes\mathbf{I}\right)\mathbf{C}_2\bar{\mathbf{U}}^{\mathrm{H}}=\bar{\mathbf{U}}\mathbf{C}_1^{\mathrm{H}}\mathbf{C}_2\bar{\mathbf{U}}^{\mathrm{H}}.\tag{66}$$

Proof: Since $\mathbf{C}_1$ and $\mathbf{C}_2$ are vector-valued block-circulant matrices, we can obtain that $\mathbf{C}_1=\left(\mathbf{F}_{\bar{N}_{\mathrm{de}}}^{\mathrm{H}}\otimes\mathbf{I}\right)\mathbf{\Lambda}_1\mathbf{F}_{\bar{N}_{\mathrm{de}}}$ and $\mathbf{C}_2=\left(\mathbf{F}_{\bar{N}_{\mathrm{de}}}^{\mathrm{H}}\otimes\mathbf{I}\right)\mathbf{\Lambda}_2\mathbf{F}_{\bar{N}_{\mathrm{de}}}$, where $\mathbf{\Lambda}_1$ and $\mathbf{\Lambda}_2$ are vector-valued block-diagonal matrices [47]. Let $\mathbf{I}_{N_{\mathrm{v}}\times\bar{N}_{\mathrm{de}}}$ denote the matrix composed of the first N_{v} rows of $\mathbf{I}_{\bar{N}_{\mathrm{de}}}$, we have

$$\begin{aligned}&\bar{\mathbf{U}}\mathbf{C}_1^{\mathrm{H}}\left(\bar{\mathbf{U}}^{\mathrm{H}}\bar{\mathbf{U}}\otimes\mathbf{I}\right)\mathbf{C}_2\bar{\mathbf{U}}^{\mathrm{H}}\\&=\mathbf{I}_{N_{\mathrm{v}}\times\bar{N}_{\mathrm{de}}}\mathbf{\Lambda}_1^{\mathrm{H}}\left(\mathbf{I}_{N_{\mathrm{v}}\times\bar{N}_{\mathrm{de}}}\mathbf{I}_{N_{\mathrm{v}}\times\bar{N}_{\mathrm{de}}}\otimes\mathbf{I}\right)\mathbf{\Lambda}_2\mathbf{I}_{N_{\mathrm{v}}\times\bar{N}_{\mathrm{de}}}^{\mathrm{H}}\\&=\mathbf{I}_{N_{\mathrm{v}}\times\bar{N}_{\mathrm{de}}}\mathbf{\Lambda}_1^{\mathrm{H}}\mathbf{\Lambda}_2\mathbf{I}_{N_{\mathrm{v}}\times\bar{N}_{\mathrm{de}}}^{\mathrm{H}}=\bar{\mathbf{U}}\mathbf{C}_1^{\mathrm{H}}\mathbf{C}_2\bar{\mathbf{U}}^{\mathrm{H}}.\end{aligned}\tag{67}$$

This completes the proof of Lemma 2. ■

Motivated by the space-frequency BBCM and (45), we define $\bar{\mathbf{W}}_u\triangleq\bar{\mathbf{V}}^{\mathrm{H}}\mathbf{W}_u$. Then MSE_u in (44) is rewritten as

$$\begin{aligned}\mathrm{MSE}_u=\mathrm{tr}\Big\{&\sigma_{\mathrm{x}}^2\mathbf{I}-\sigma_{\mathrm{x}}^2\bar{\mathbf{W}}_u^{\mathrm{H}}\left(\bar{\mathbf{U}}\otimes\mathbf{I}\right)\hat{\mathbf{H}}_u^{\mathrm{SF}}\bar{\mathbf{U}}^{\mathrm{H}}\\&-\sigma_{\mathrm{x}}^2\bar{\mathbf{U}}\big(\hat{\mathbf{H}}_u^{\mathrm{SF}}\big)^{\mathrm{H}}\left(\bar{\mathbf{U}}^{\mathrm{H}}\otimes\mathbf{I}\right)\bar{\mathbf{W}}_u+\sigma_{\mathrm{z}}^2\mathbf{W}_u^{\mathrm{H}}\mathbf{W}_u\\&+\sigma_{\mathrm{x}}^2\sum_{u'=0}^{U-1}\bar{\mathbf{W}}_u^{\mathrm{H}}\left(\bar{\mathbf{U}}\otimes\mathbf{I}\right)\hat{\mathbf{H}}_{u'}^{\mathrm{SF}}\bar{\mathbf{U}}^{\mathrm{H}}\bar{\mathbf{U}}\big(\hat{\mathbf{H}}_{u'}^{\mathrm{SF}}\big)^{\mathrm{H}}\\&\times\left(\bar{\mathbf{U}}^{\mathrm{H}}\otimes\mathbf{I}\right)\bar{\mathbf{W}}_u\Big\}.\end{aligned}\tag{68}$$

$$\tilde{\mathbf{H}}_{u,k,\tilde{n}_{\mathrm{an}}}^{\mathrm{S}}\left(\tilde{\mathbf{H}}_{u,k,\tilde{n}_{\mathrm{an}}}^{\mathrm{S}}\right)^{\mathrm{H}}+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}=\begin{bmatrix}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}^{\mathrm{H}}+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2} & \left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}^{\mathrm{H}}\\ \left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}^{\mathrm{H}} & \left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}^{\mathrm{H}}+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\end{bmatrix}\tag{64}$$

$$\left(\tilde{\mathbf{H}}_{u,k,\tilde{n}_{\mathrm{an}}}^{\mathrm{S}}\left(\tilde{\mathbf{H}}_{u,k,\tilde{n}_{\mathrm{an}}}^{\mathrm{S}}\right)^{\mathrm{H}}+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\right)^{-1}=\begin{bmatrix}\mathbf{A}_1 & \mathbf{A}_2\\ \mathbf{A}_3 & \left(\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}\left(\frac{\sigma_{\mathrm{x}}^2}{\sigma_{\mathrm{z}}^2}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}^{\mathrm{H}}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}+\mathbf{I}\right)^{-1}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}^{\mathrm{H}}+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\right)^{-1}\end{bmatrix},\tag{65}$$

$$\left(\tilde{\mathbf{H}}_{u,k,\tilde{n}_{\mathrm{an}}}^{\mathrm{S}}\left(\tilde{\mathbf{H}}_{u,k,\tilde{n}_{\mathrm{an}}}^{\mathrm{S}}\right)^{\mathrm{H}}+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\right)^{-1}=\begin{bmatrix}\left(\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\tilde{n}_{\mathrm{an}},:}^{\mathrm{H}}+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\right)^{-1} & \mathbf{0}\\ \mathbf{0} & \left(\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}\left[\tilde{\mathbf{H}}_{k}^{\mathrm{S}}\right]_{\mathcal{N}_{u,1}^{\mathrm{an}},:}^{\mathrm{H}}+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\right)^{-1}\end{bmatrix}\tag{66}$$

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Next, it is reasonable to assume that $\bar{\mathbf{W}}_u = \left(\bar{\mathbf{U}} \otimes \mathbf{I}\right) \tilde{\mathbf{W}}_u \bar{\mathbf{U}}^{\mathrm{H}}$ according to Lemma 1 since $\left(\bar{\mathbf{U}}\bar{\mathbf{U}}^{\dagger} \otimes \mathbf{I}\right) \bar{\mathbf{W}}_u \left(\bar{\mathbf{U}}^{\mathrm{H}}\right)^{\dagger} \bar{\mathbf{U}}^{\mathrm{H}} = \bar{\mathbf{W}}_u$ is always satisfied and thus $\tilde{\mathbf{W}}_u$ always exists. Since $\mathbf{W}_u$ is a vector-valued block-diagonal matrix, $\bar{\mathbf{W}}_u$ should also be a vector-valued block-diagonal matrix, and thus $\tilde{\mathbf{W}}_u$ is a vector-valued block-circulant matrix [47]. Then MSE_u in (68) becomes

$$\begin{aligned}\mathrm{MSE}_u = \mathrm{tr}\Big\{ & \bar{\mathbf{U}}\Big(\sigma_{\mathrm{x}}^2\mathbf{I} - \sigma_{\mathrm{x}}^2 \tilde{\mathbf{W}}_u^{\mathrm{H}} \left(\bar{\mathbf{U}}^{\mathrm{H}}\bar{\mathbf{U}} \otimes \mathbf{I}\right) \hat{\mathbf{H}}_u^{\mathrm{SF}} \\ & - \sigma_{\mathrm{x}}^2 \left(\hat{\mathbf{H}}_u^{\mathrm{SF}}\right)^{\mathrm{H}} \left(\bar{\mathbf{U}}^{\mathrm{H}}\bar{\mathbf{U}} \otimes \mathbf{I}\right) \tilde{\mathbf{W}}_u + \sigma_{\mathrm{z}}^2 \mathbf{W}_u^{\mathrm{H}} \mathbf{W}_u \\ & + \sigma_{\mathrm{x}}^2 \sum_{u'=0}^{U-1} \tilde{\mathbf{W}}_u^{\mathrm{H}} \left(\bar{\mathbf{U}}^{\mathrm{H}}\bar{\mathbf{U}} \otimes \mathbf{I}\right) \hat{\mathbf{H}}_{u'}^{\mathrm{SF}} \bar{\mathbf{U}}^{\mathrm{H}} \bar{\mathbf{U}} \left(\hat{\mathbf{H}}_{u'}^{\mathrm{SF}}\right)^{\mathrm{H}} \\ & \times \left(\bar{\mathbf{U}}^{\mathrm{H}}\bar{\mathbf{U}} \otimes \mathbf{I}\right) \tilde{\mathbf{W}}_u \Big) \bar{\mathbf{U}}^{\mathrm{H}} \Big\}. \end{aligned} \tag{69}$$

According to Lemma 2, MSE_u in (69) can be rewritten as

$$\begin{aligned}\mathrm{MSE}_u = \mathrm{tr}\Big\{ & \bar{\mathbf{U}}\Big(\sigma_{\mathrm{x}}^2\mathbf{I} - \sigma_{\mathrm{x}}^2 \tilde{\mathbf{W}}_u^{\mathrm{H}} \hat{\mathbf{H}}_u^{\mathrm{SF}} - \sigma_{\mathrm{x}}^2 \left(\hat{\mathbf{H}}_u^{\mathrm{SF}}\right)^{\mathrm{H}} \tilde{\mathbf{W}}_u \\ & + \sigma_{\mathrm{x}}^2 \sum_{u'=0}^{U-1} \tilde{\mathbf{W}}_u^{\mathrm{H}} \hat{\mathbf{H}}_{u'}^{\mathrm{SF}} \left(\hat{\mathbf{H}}_{u'}^{\mathrm{SF}}\right)^{\mathrm{H}} \tilde{\mathbf{W}}_u \Big) \bar{\mathbf{U}}^{\mathrm{H}} \\ & + \sigma_{\mathrm{z}}^2 \mathbf{W}_u^{\mathrm{H}} \mathbf{W}_u \Big\}. \end{aligned} \tag{70}$$

In (70), we can turn to optimizing $\tilde{\mathbf{W}}_u$ instead of $\mathbf{W}_u$ to minimize the MSE. Although $\mathbf{W}_u$ is still involved in the term $\sigma_{\mathrm{z}}^2 \mathbf{W}_u^{\mathrm{H}} \mathbf{W}_u$, it can be obtained when $\tilde{\mathbf{W}}_u$ is given according to the relation $\bar{\mathbf{W}}_u = \bar{\mathbf{V}}^{\mathrm{H}} \mathbf{W}_u = \left(\bar{\mathbf{U}} \otimes \mathbf{I}\right) \tilde{\mathbf{W}}_u \bar{\mathbf{U}}^{\mathrm{H}}$. Hence, $\tilde{\mathbf{W}}_u$ needs to satisfy the constraint $\bar{\mathbf{V}}^{\mathrm{H}} \left(\bar{\mathbf{V}}^{\mathrm{H}}\right)^{\dagger} \left(\bar{\mathbf{U}} \otimes \mathbf{I}\right) \tilde{\mathbf{W}}_u \bar{\mathbf{U}}^{\mathrm{H}} = \left(\bar{\mathbf{U}} \otimes \mathbf{I}\right) \tilde{\mathbf{W}}_u \bar{\mathbf{U}}^{\mathrm{H}}$ to guarantee the existence of $\mathbf{W}_u$ according to Lemma 1.

Note that for a given optimal solution $\tilde{\mathbf{W}}_u^{\mathrm{op}}$, the corresponding $\mathbf{W}_u^{\mathrm{op}}$ should have the minimal $\mathrm{tr}\left(\left(\mathbf{W}_u^{\mathrm{op}}\right)^{\mathrm{H}} \mathbf{W}_u^{\mathrm{op}}\right)$, and we can obtain the corresponding MSE_u in (70). Otherwise, we will have a $\mathbf{W}_u^{\mathrm{op}}$ with lower $\mathrm{tr}\left(\mathbf{W}_u^{\mathrm{H}} \mathbf{W}_u\right)$, which leads to a lower MSE_u and contradicts the optimality. Therefore, we could formulate the following optimization problem

$$\begin{aligned} & \mathbf{W}_u^{\mathrm{op}} = \arg\min_{\mathbf{W}_u} \ \mathrm{tr}\left(\mathbf{W}_u^{\mathrm{H}} \mathbf{W}_u\right), \\ & \text{s.t.} \ \left(\bar{\mathbf{U}} \otimes \mathbf{I}\right) \tilde{\mathbf{W}}_u \bar{\mathbf{U}}^{\mathrm{H}} = \bar{\mathbf{V}}^{\mathrm{H}} \mathbf{W}_u. \end{aligned} \tag{71}$$

According to the vector-valued block-diagonal structure of $\mathbf{W}_u^{\mathrm{op}}$, (71) is equivalent to

$$\begin{aligned} & \mathrm{vec}\left(\mathbf{W}_u^{\mathrm{op}}\right) = \arg\min_{\mathrm{vec}(\mathbf{W}_u)} \ \mathrm{vec}(\mathbf{W}_u)^{\mathrm{H}} \mathrm{vec}\left(\mathbf{W}_u\right), \\ & \text{s.t.} \ \left(\bar{\mathbf{U}}^* \otimes \bar{\mathbf{U}} \otimes \mathbf{I}\right) \mathrm{vec}\left(\tilde{\mathbf{W}}_u^{\mathrm{op}}\right) \\ & \qquad = \left(\mathbf{I} \otimes \bar{\mathbf{V}}^{\mathrm{H}}\right) \mathrm{vec}\left(\mathbf{W}_u\right), \end{aligned} \tag{72}$$

whose optimal solution can be obtained as

$$\mathrm{vec}\left(\mathbf{W}_u^{\mathrm{op}}\right) = \left(\bar{\mathbf{U}}^* \otimes \bar{\mathbf{V}}\left(\bar{\mathbf{V}}^{\mathrm{H}}\bar{\mathbf{V}}\right)^{\dagger} \left(\bar{\mathbf{U}} \otimes \mathbf{I}\right)\right) \mathrm{vec}\left(\tilde{\mathbf{W}}_u^{\mathrm{op}}\right). \tag{73}$$

Thus we have

$$\mathbf{W}_u^{\mathrm{op}} = \bar{\mathbf{V}}\left(\bar{\mathbf{V}}^{\mathrm{H}}\bar{\mathbf{V}}\right)^{\dagger} \left(\bar{\mathbf{U}} \otimes \mathbf{I}\right) \tilde{\mathbf{W}}_u^{\mathrm{op}} \bar{\mathbf{U}}^{\mathrm{H}}. \tag{74}$$

Then, according to (74), we replace $\mathbf{W}_u$ with $\bar{\mathbf{V}}\left(\bar{\mathbf{V}}^{\mathrm{H}}\bar{\mathbf{V}}\right)^{\dagger} \left(\bar{\mathbf{U}} \otimes \mathbf{I}\right) \tilde{\mathbf{W}}_u \bar{\mathbf{U}}^{\mathrm{H}}$ in (70) and obtain the objective function in (47). This completes the proof.

Appendix E
Proof of Theorem 5

The objective function of problem (49) can be rewritten as

$$\begin{aligned} \mathrm{tr}\Big\{ & \bar{\mathbf{U}}\Big(\sigma_{\mathrm{x}}^2 \left(\tilde{\mathbf{W}}_u - \mathbf{T}^{-1}\hat{\mathbf{H}}_u^{\mathrm{SF}}\right)^{\mathrm{H}} \mathbf{T} \left(\tilde{\mathbf{W}}_u - \mathbf{T}^{-1}\hat{\mathbf{H}}_u^{\mathrm{SF}}\right) \\ & + \sigma_{\mathrm{x}}^2 \Big(\mathbf{I} - \left(\hat{\mathbf{H}}_u^{\mathrm{SF}}\right)^{\mathrm{H}} \mathbf{T}^{-1} \hat{\mathbf{H}}_u^{\mathrm{SF}}\Big)\Big) \bar{\mathbf{U}}^{\mathrm{H}} \Big\}, \end{aligned} \tag{75}$$

where $\mathbf{T} = \sum_{u'=0}^{U-1} \hat{\mathbf{H}}_{u'}^{\mathrm{SF}} \left(\hat{\mathbf{H}}_{u'}^{\mathrm{SF}}\right)^{\mathrm{H}} + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}$. It can be found that to minimize the objective function, the optimal $\tilde{\mathbf{W}}_u^{\mathrm{op},\infty}$ is given by

$$\tilde{\mathbf{W}}_u^{\mathrm{op},\infty} = \mathbf{T}^{-1}\hat{\mathbf{H}}_u^{\mathrm{SF}} = \Big(\sum_{u'=0}^{U-1} \hat{\mathbf{H}}_{u'}^{\mathrm{SF}} \left(\hat{\mathbf{H}}_{u'}^{\mathrm{SF}}\right)^{\mathrm{H}} + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\Big)^{-1} \hat{\mathbf{H}}_u^{\mathrm{SF}}. \tag{76}$$

Moreover, we have

$$\begin{aligned} & \Big(\sum_{u'=0}^{U-1} \hat{\mathbf{H}}_{u'}^{\mathrm{SF}} \left(\hat{\mathbf{H}}_{u'}^{\mathrm{SF}}\right)^{\mathrm{H}} + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\Big) \hat{\mathbf{H}}_u^{\mathrm{SF}} \\ & \overset{(a)}{=} \Big(\hat{\mathbf{H}}_u^{\mathrm{SF}} \left(\hat{\mathbf{H}}_u^{\mathrm{SF}}\right)^{\mathrm{H}} + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\Big) \hat{\mathbf{H}}_u^{\mathrm{SF}} \\ & = \hat{\mathbf{H}}_u^{\mathrm{SF}} \Big(\left(\hat{\mathbf{H}}_u^{\mathrm{SF}}\right)^{\mathrm{H}} \hat{\mathbf{H}}_u^{\mathrm{SF}} + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\Big) \\ & \Rightarrow \Big(\sum_{u'=0}^{U-1} \hat{\mathbf{H}}_{u'}^{\mathrm{SF}} \left(\hat{\mathbf{H}}_{u'}^{\mathrm{SF}}\right)^{\mathrm{H}} + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\Big)^{-1} \hat{\mathbf{H}}_u^{\mathrm{SF}} \\ & = \hat{\mathbf{H}}_u^{\mathrm{SF}} \Big(\left(\hat{\mathbf{H}}_u^{\mathrm{SF}}\right)^{\mathrm{H}} \hat{\mathbf{H}}_u^{\mathrm{SF}} + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\Big)^{-1}, \end{aligned} \tag{77}$$

where (a) follows from the condition $\forall u, u', \mathcal{N}_u^{\mathrm{an}} \cap \mathcal{N}_{u'}^{\mathrm{an}} = \emptyset$. Note that due to the condtion $\forall u, n_{\mathrm{an}}, \left|\mathcal{N}_{u,n_{\mathrm{an}}}^{\mathrm{de}}\right| \leq 1$, we have

$$\Big(\left(\hat{\mathbf{H}}_u^{\mathrm{SF}}\right)^{\mathrm{H}} \hat{\mathbf{H}}_u^{\mathrm{SF}} + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\mathbf{I}\Big)^{-1} = \Big(\left\|\hat{\mathbf{h}}_u^{\mathrm{SF}}\right\|^2 + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\Big)^{-1} \mathbf{I}. \tag{78}$$

By combining (76), (77) and (78), we obtain that $\tilde{\mathbf{W}}_u^{\mathrm{op},\infty} = \Big(\left\|\hat{\mathbf{h}}_u^{\mathrm{SF}}\right\|^2 + \frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{x}}^2}\Big)^{-1} \hat{\mathbf{H}}_u^{\mathrm{SF}}$. This completes the proof.

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